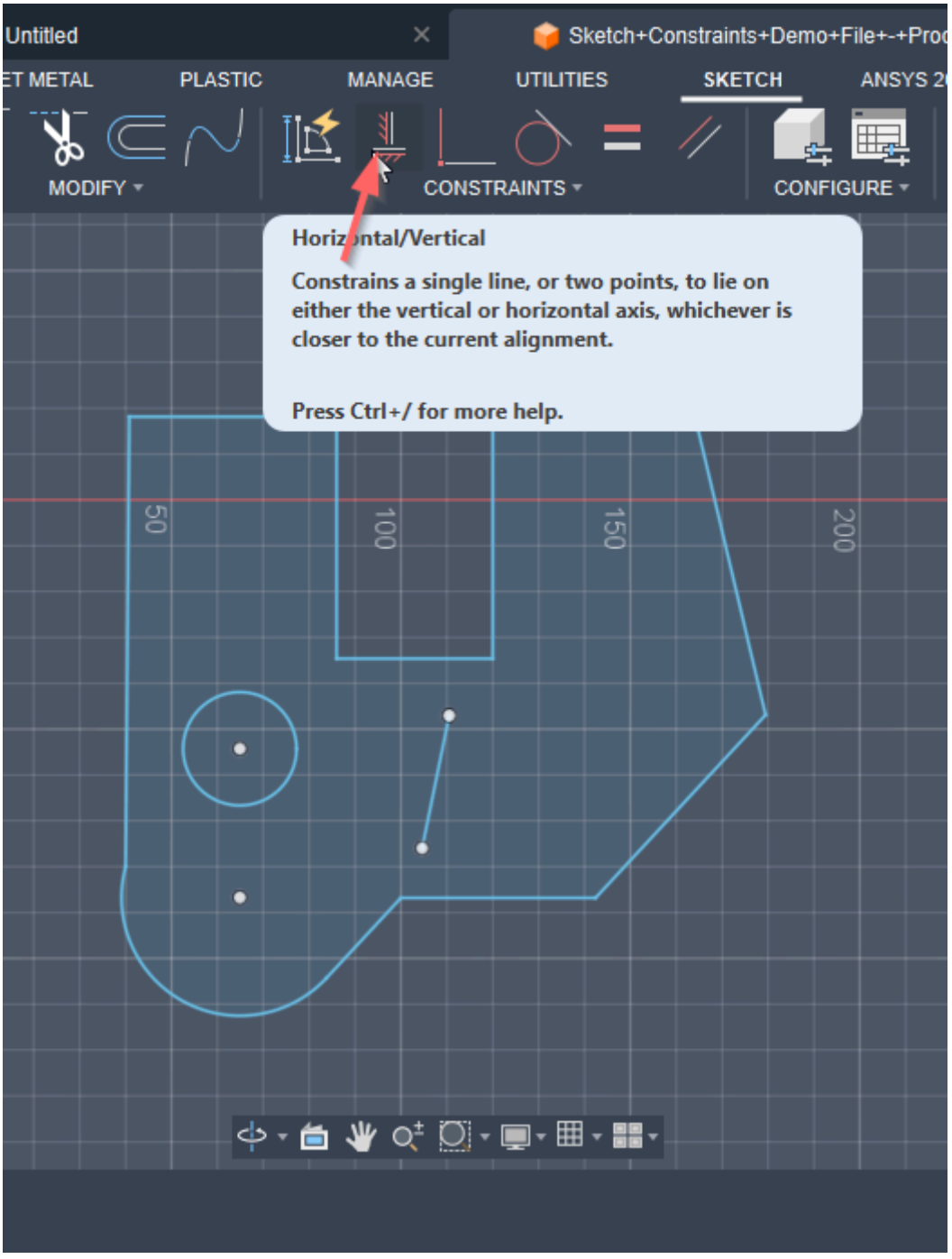
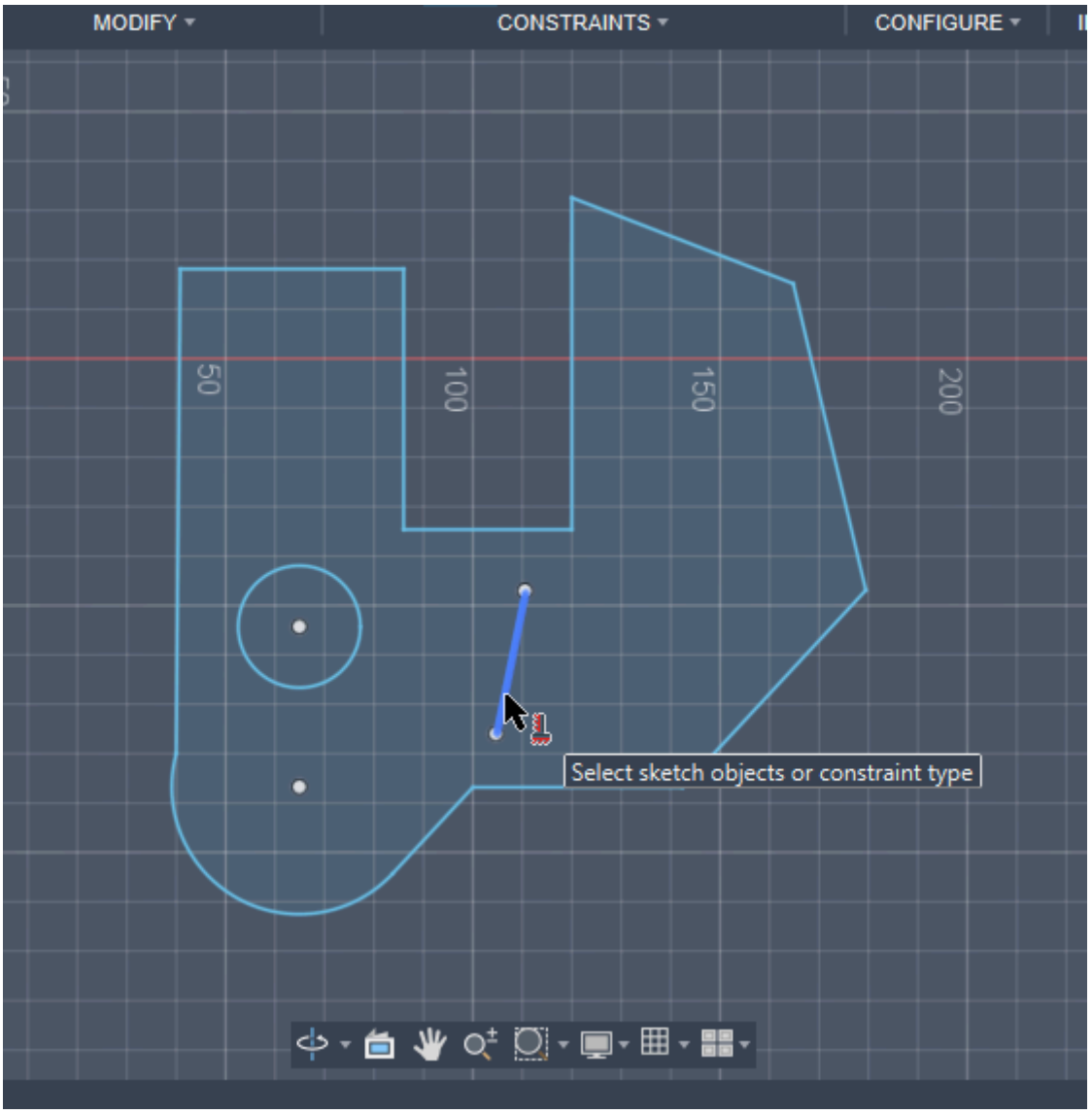
Sketch Constraints

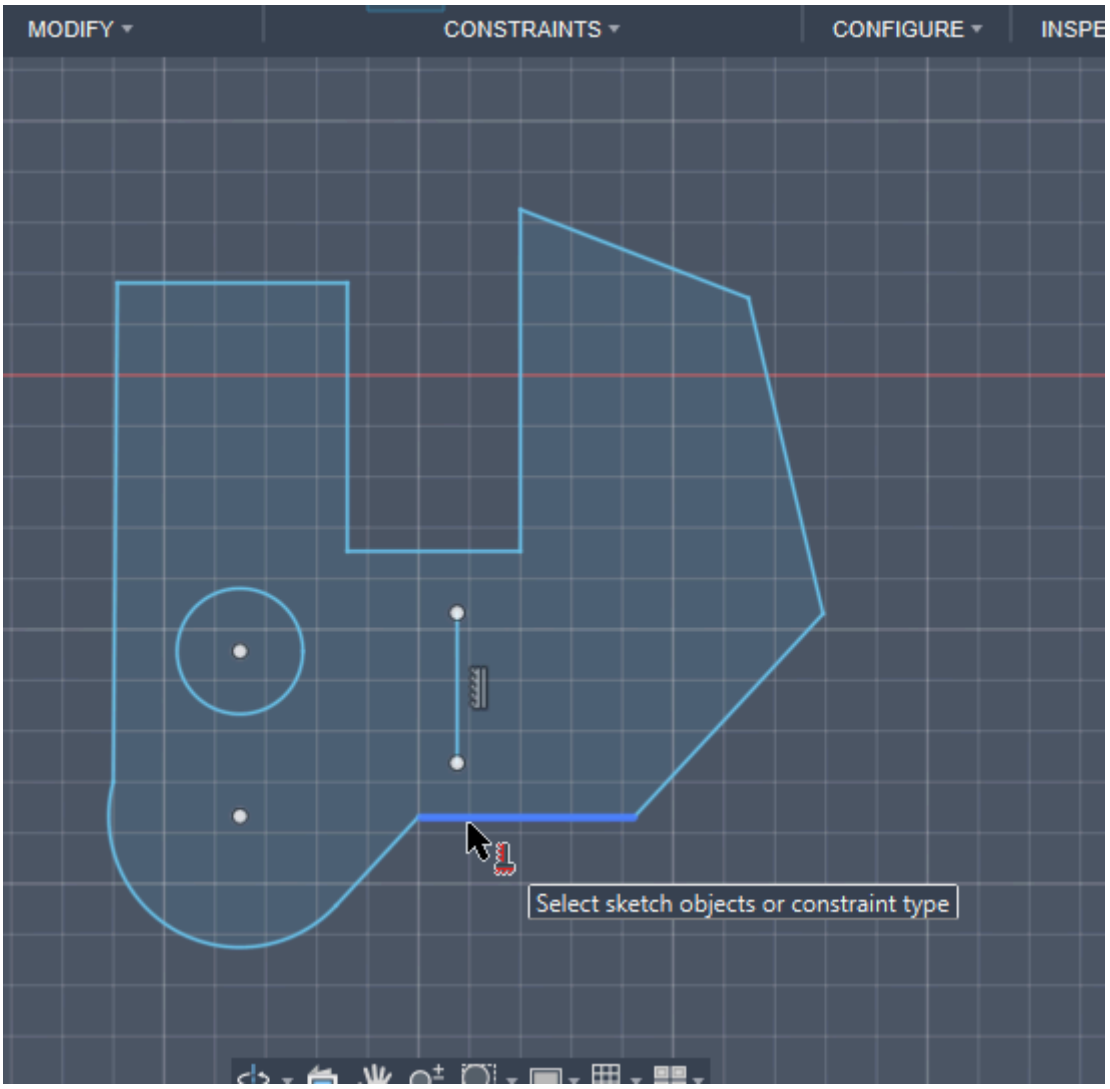
Horizontal / Vertical Constraint



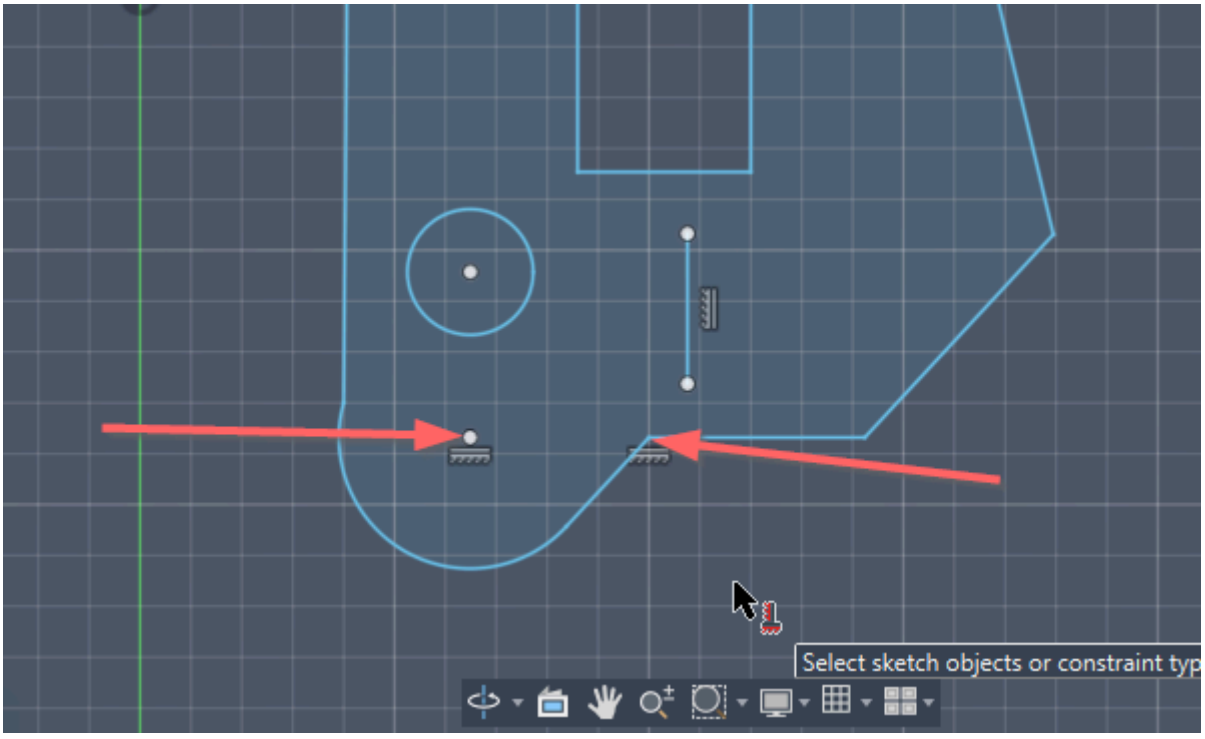
Horizontal/vertical constraint tool.



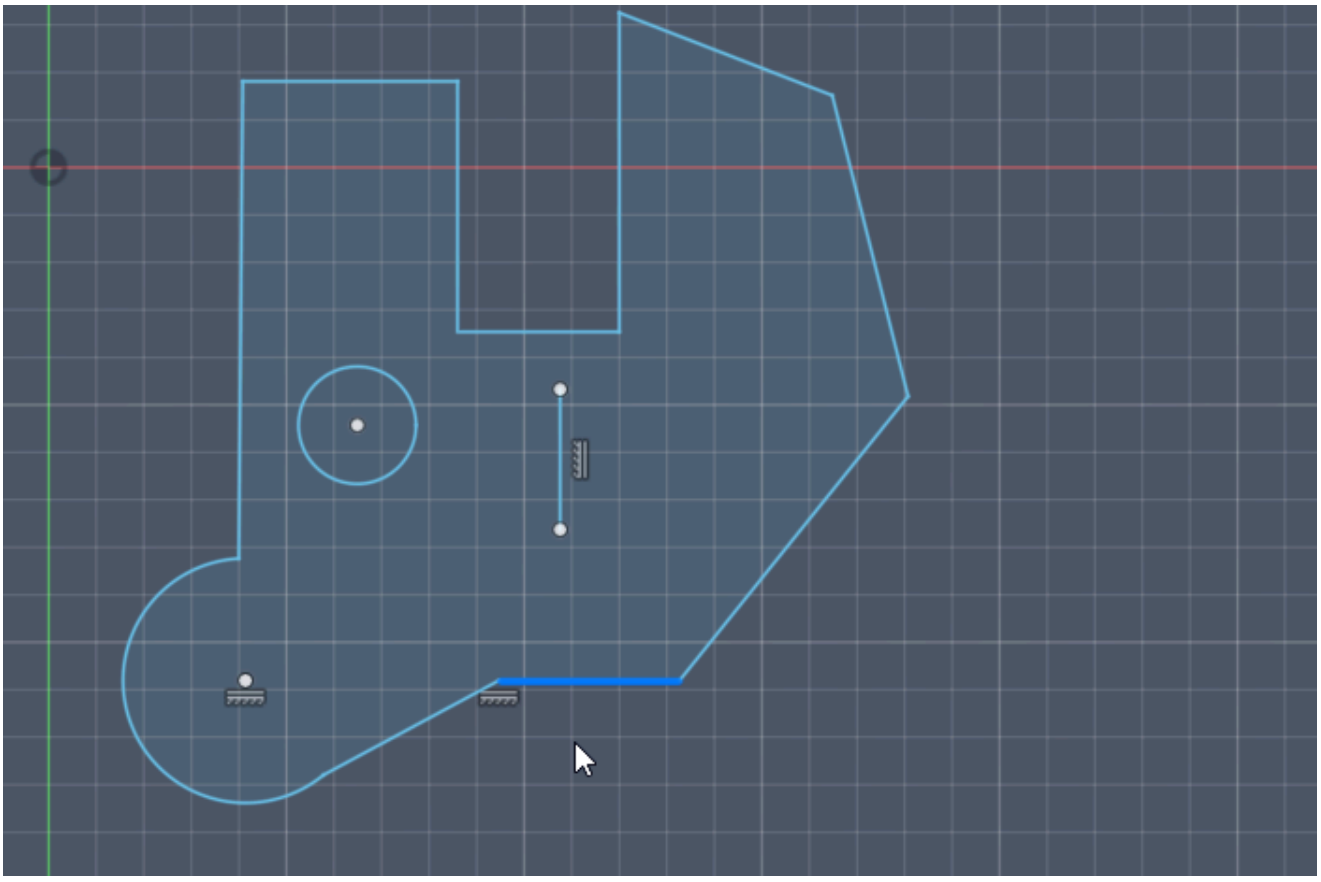
Click this line.



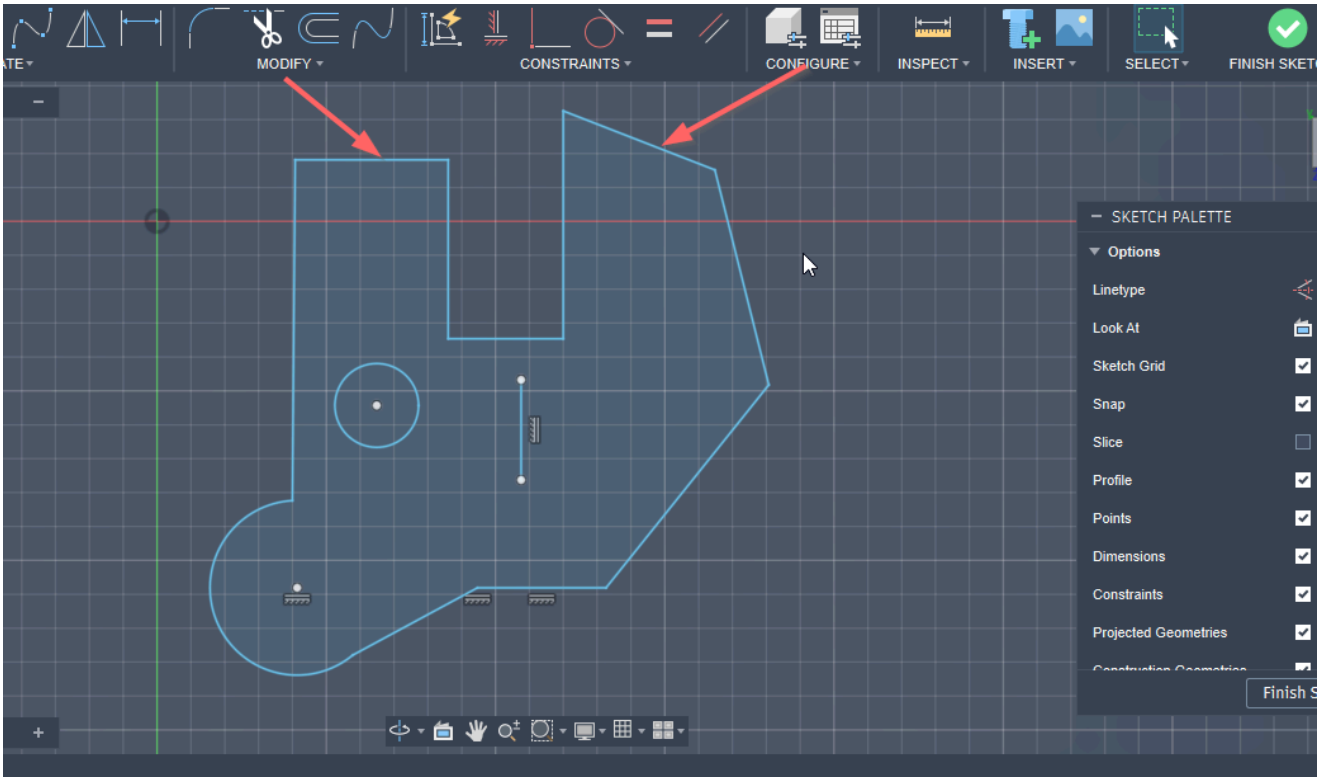
The line becomes straight.



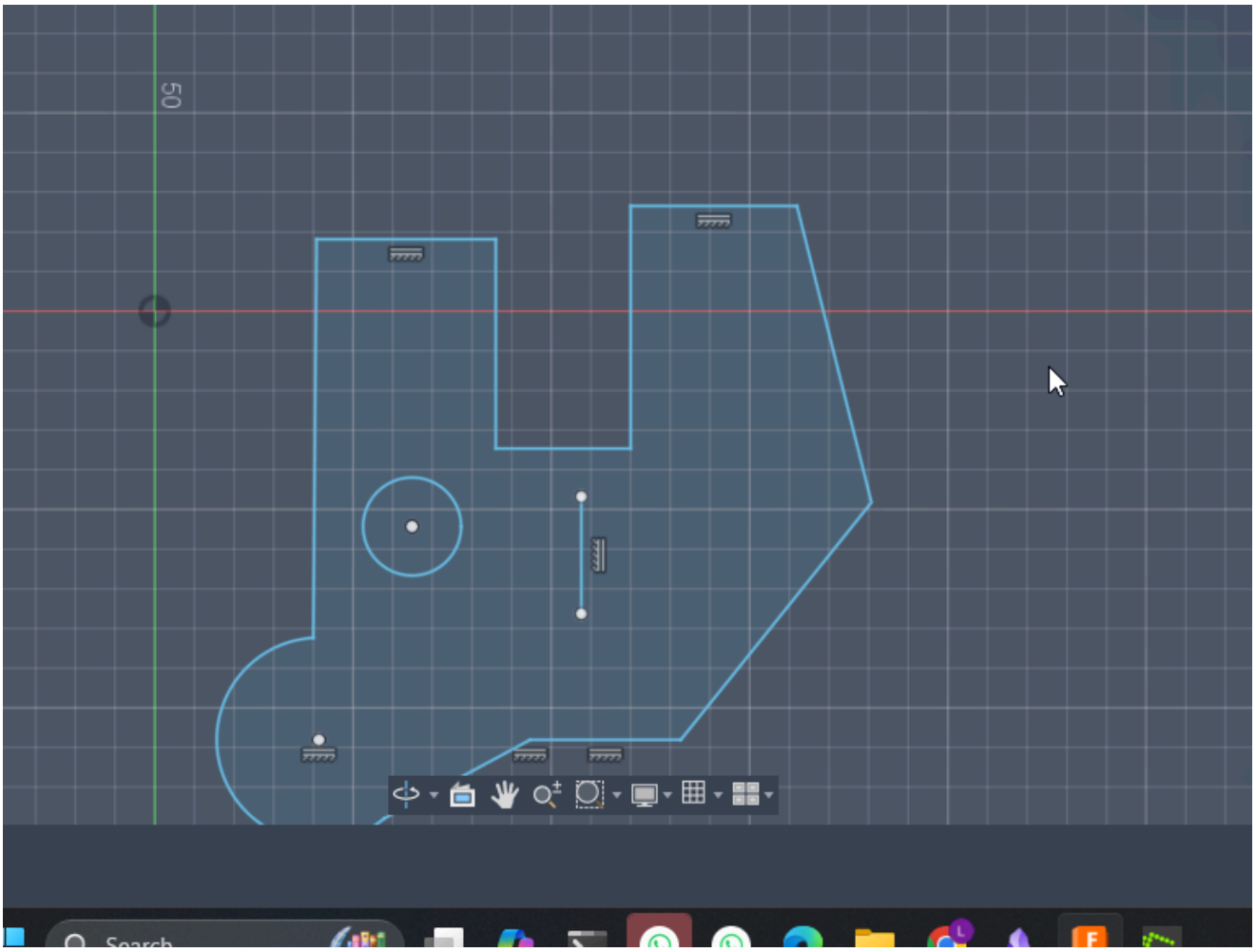
Used this constraint so the circle/arc stays on this axis.



After moving the line, the arc also stays aligned on the same axis.

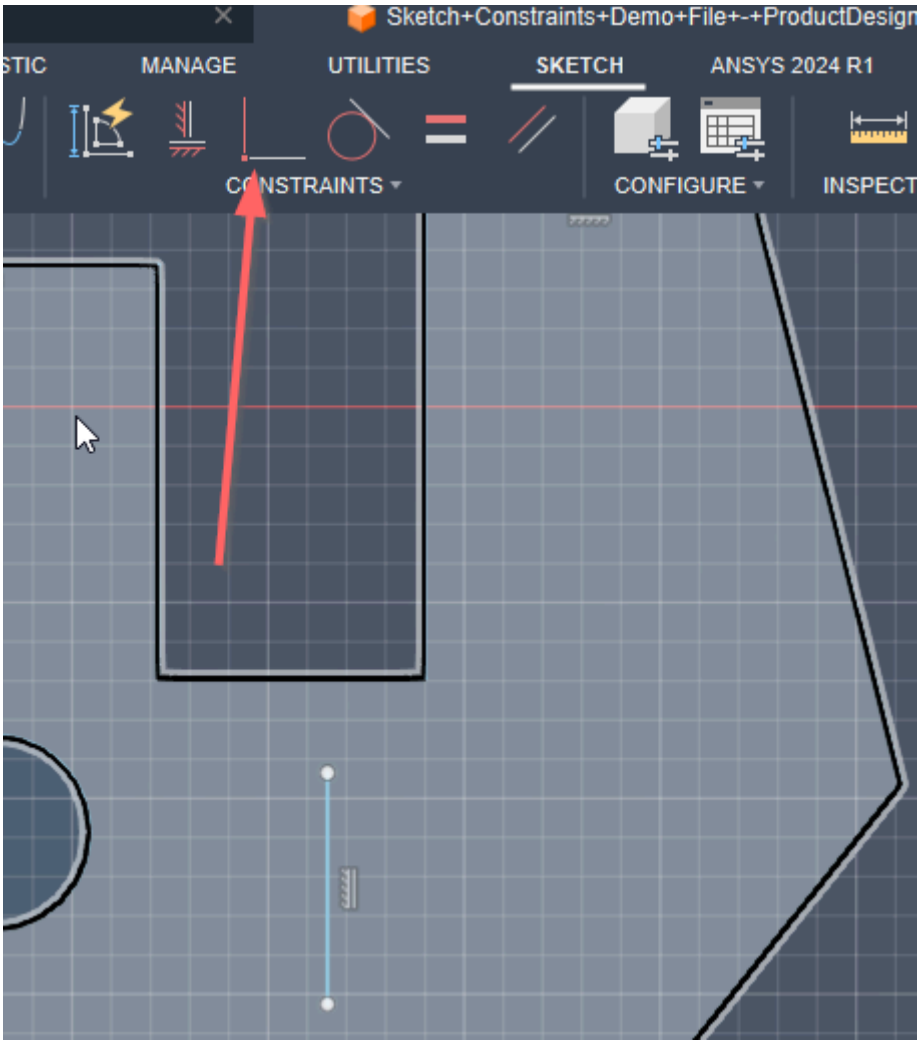


If we apply horizontal/vertical to both lines…

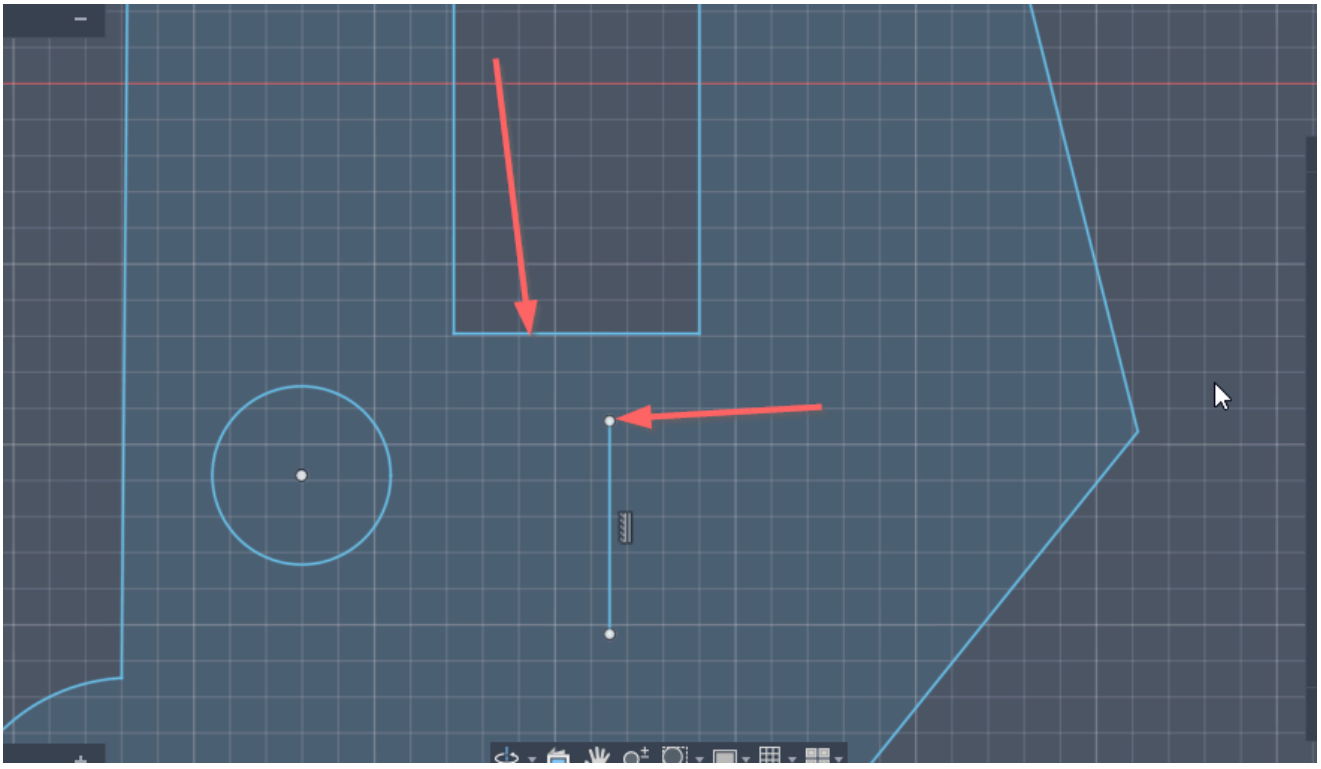


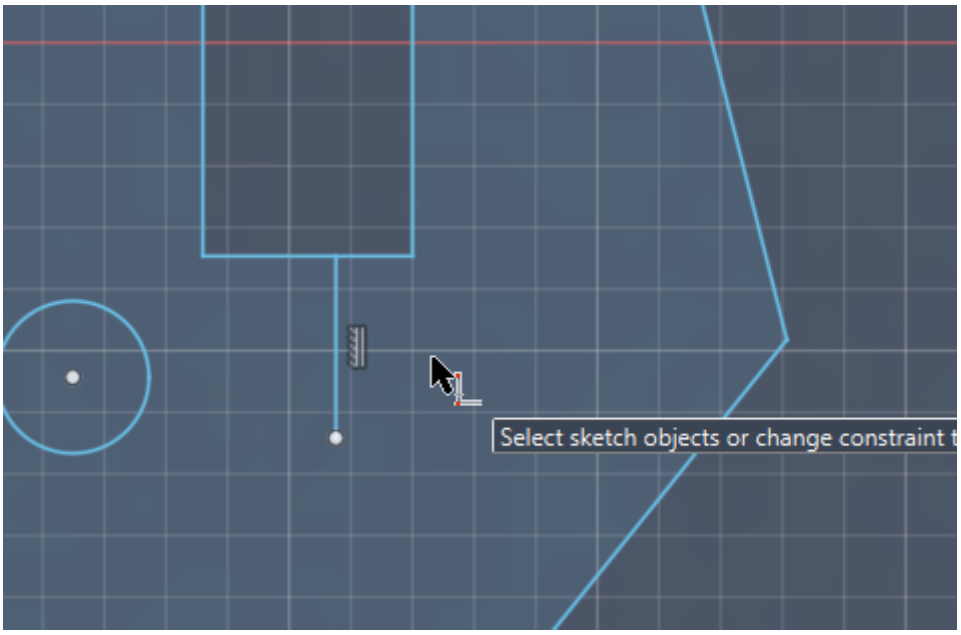
Both lines are forced straight.

Coincident Constraint

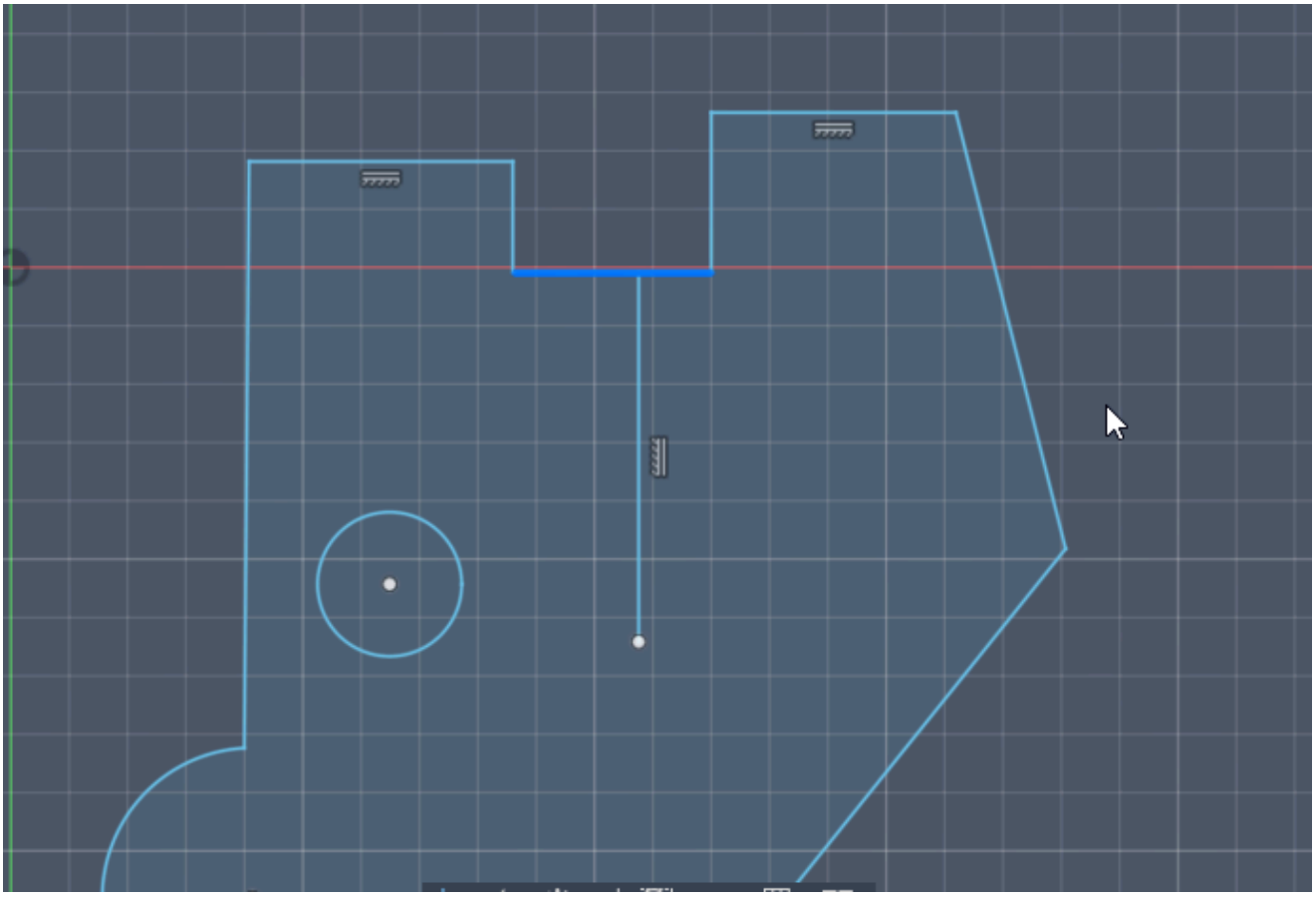


Coincident tool.

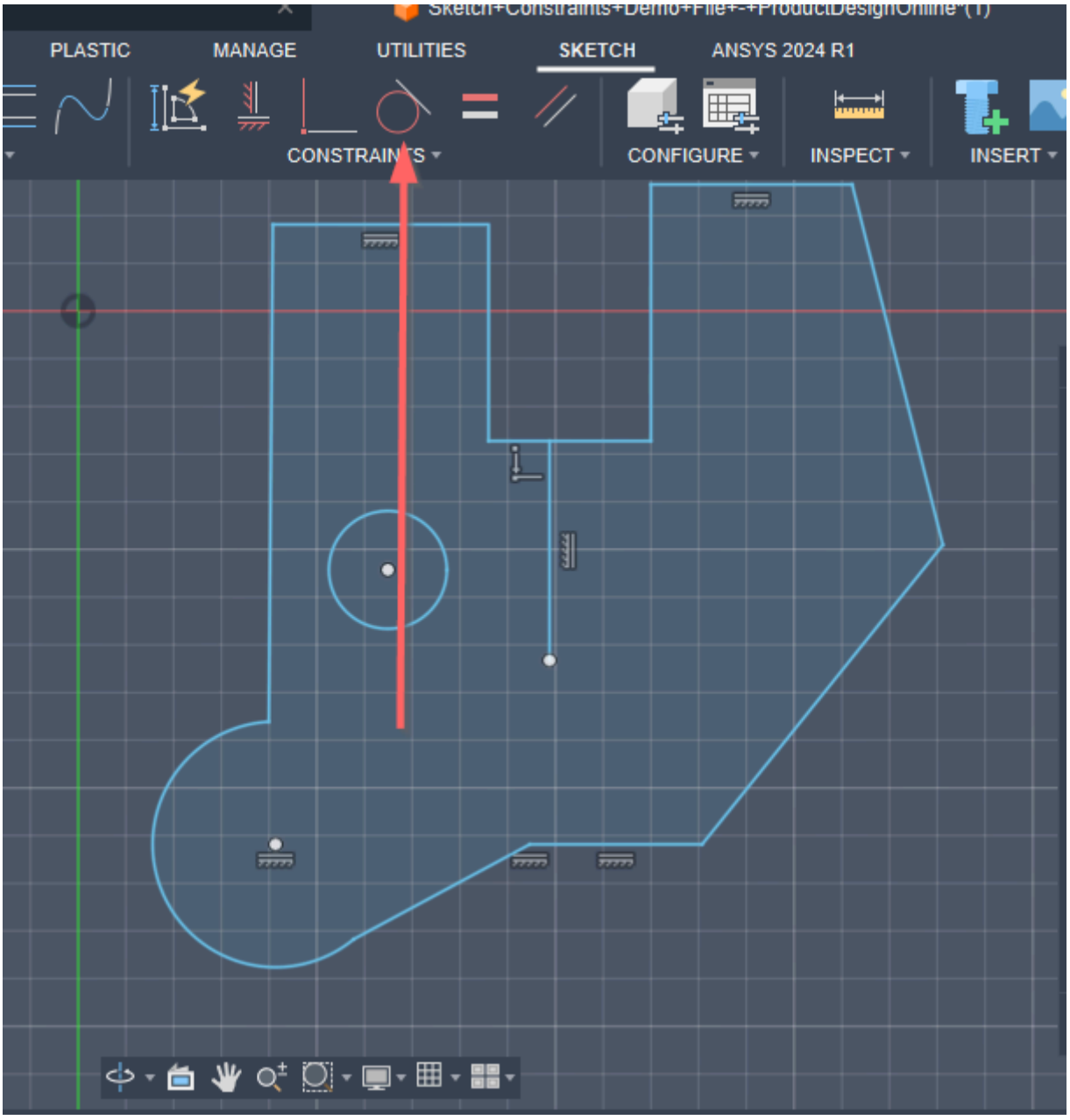

Join these two using coincident.



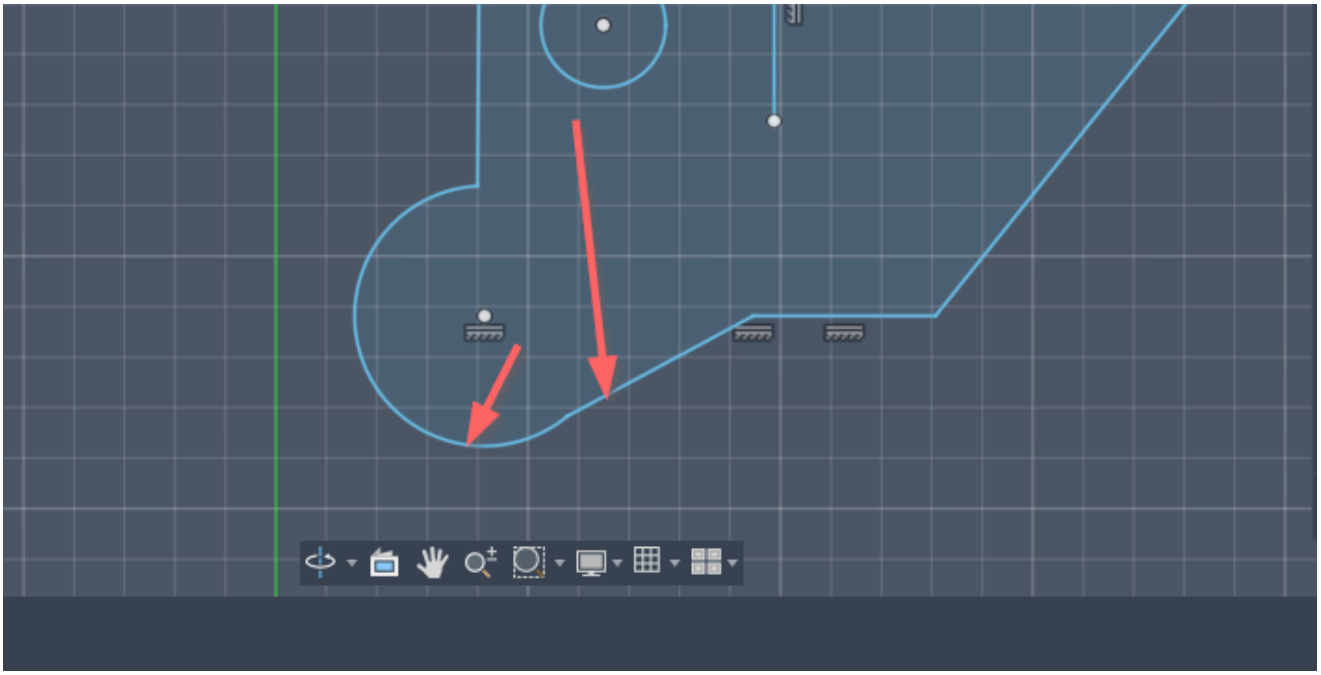
Joined.

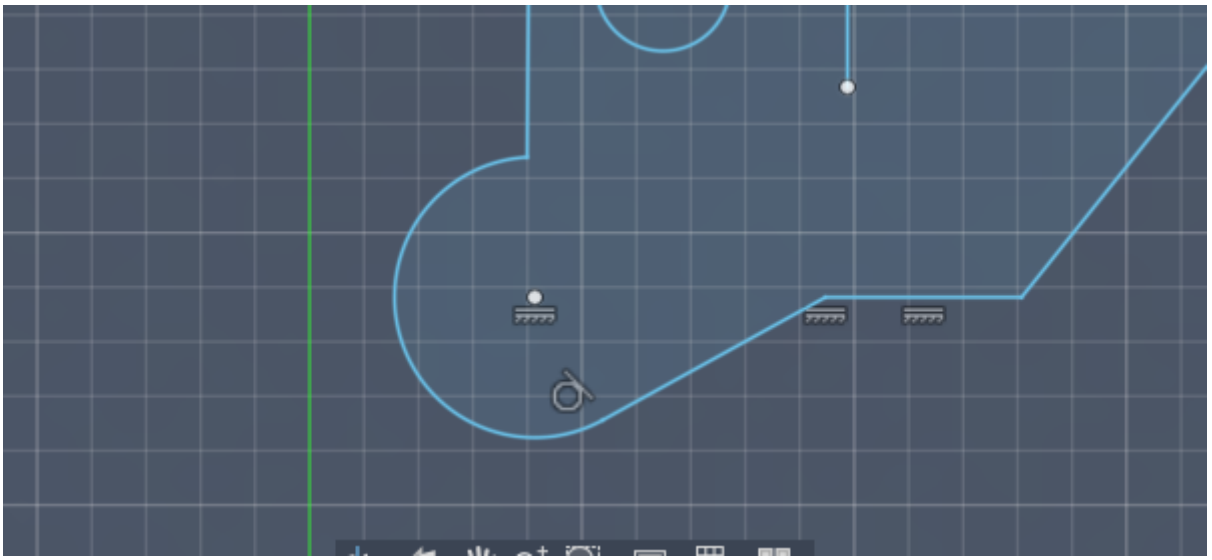

If one is moved, the other moves with it.

Tangent Constraint

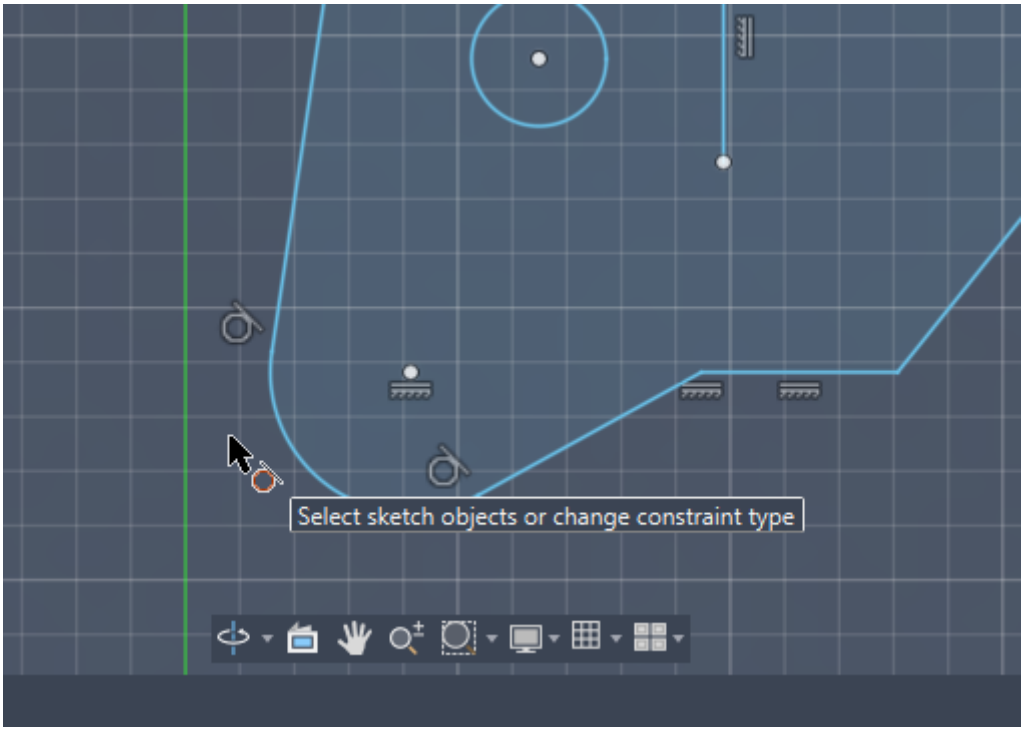


Tangent tool.


Select both to create a smooth transition.

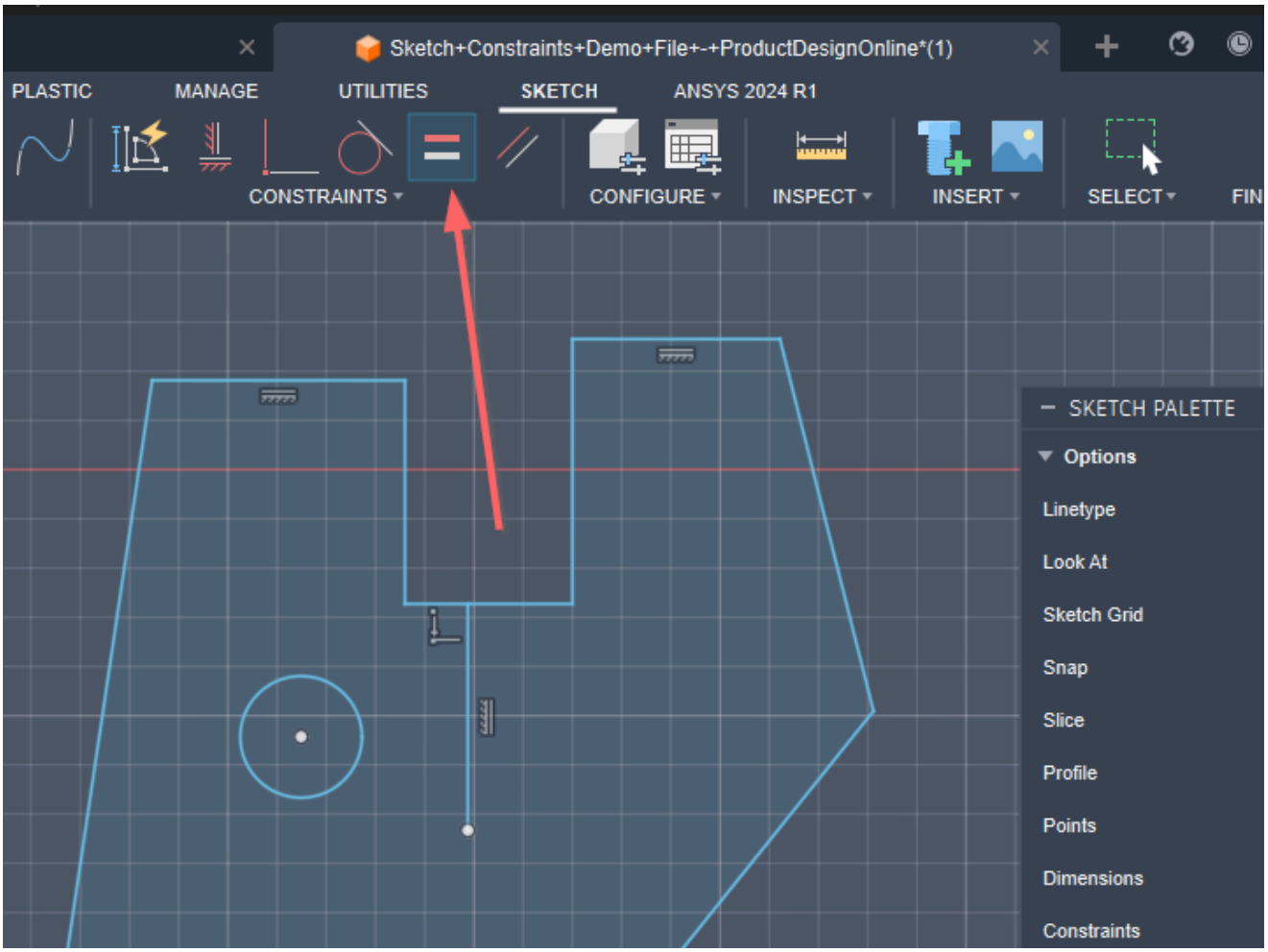


Result.

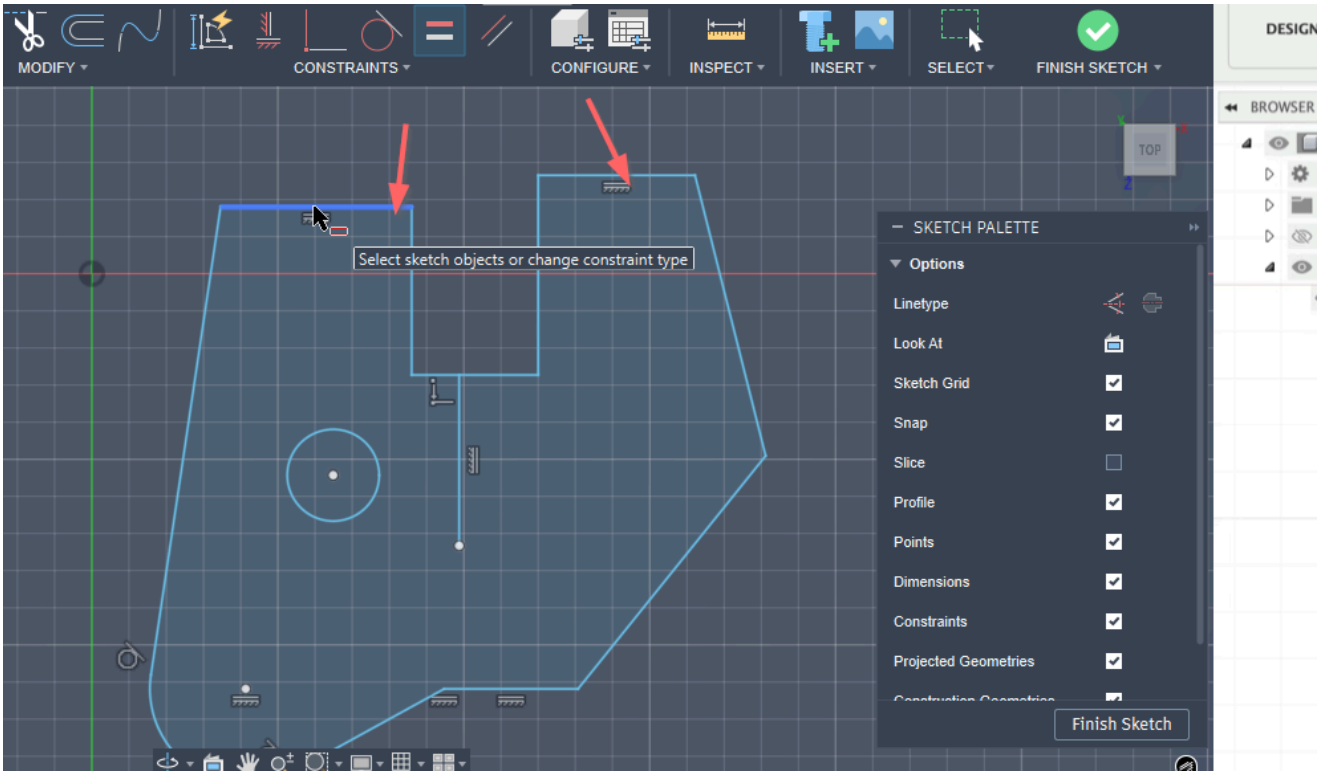


Example 2.

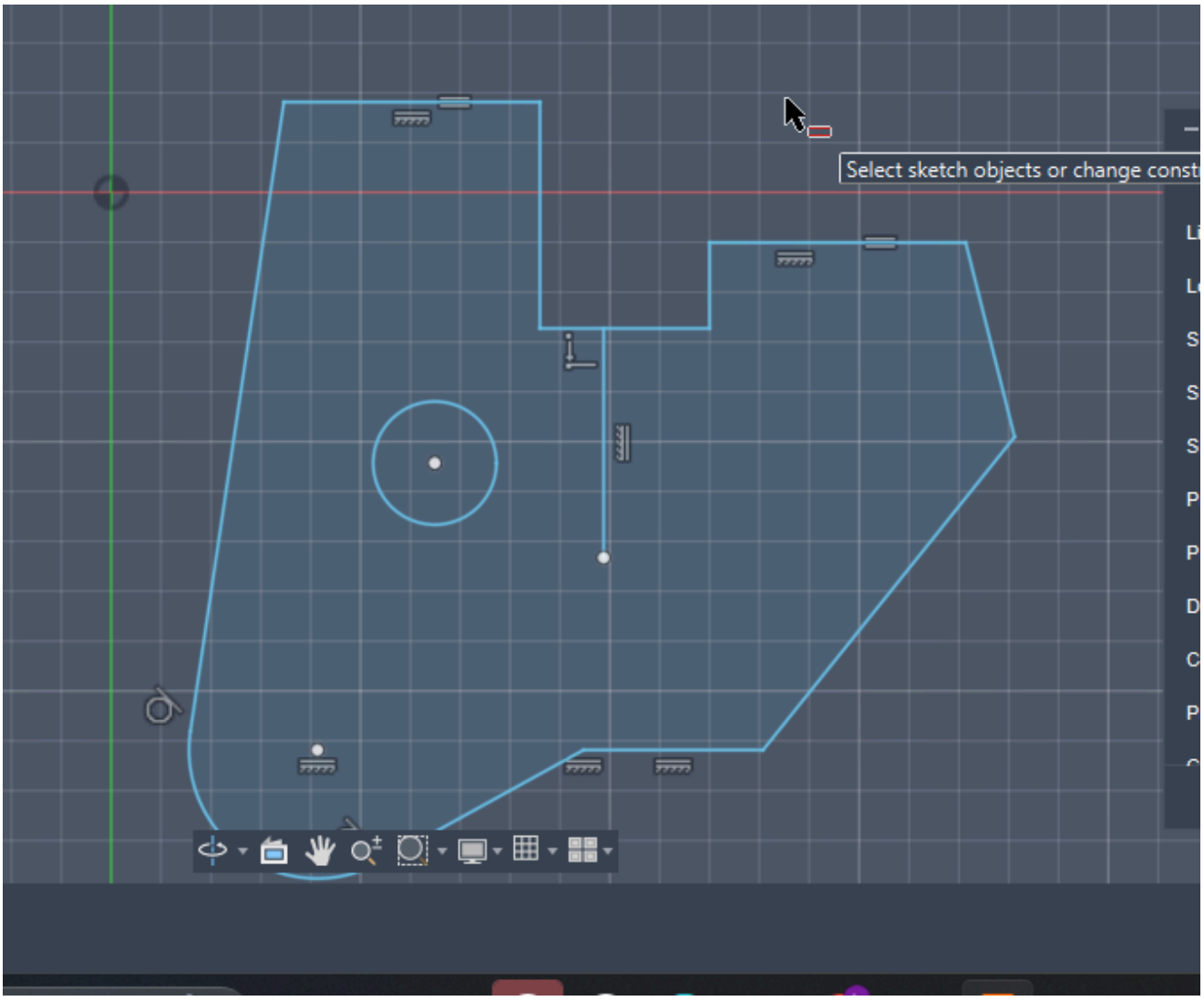
Equal Constraint



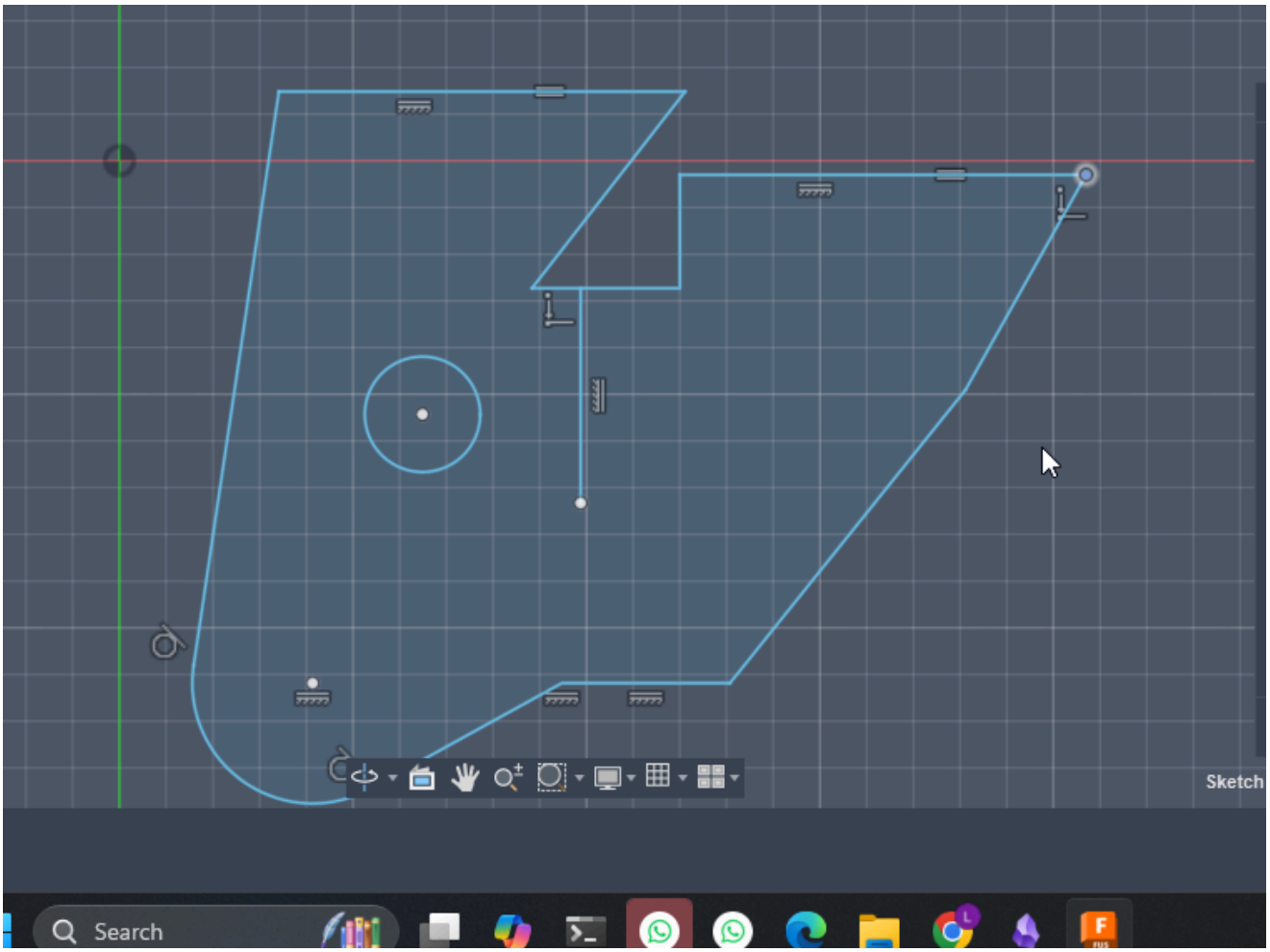
Equal tool.



Select both lines.

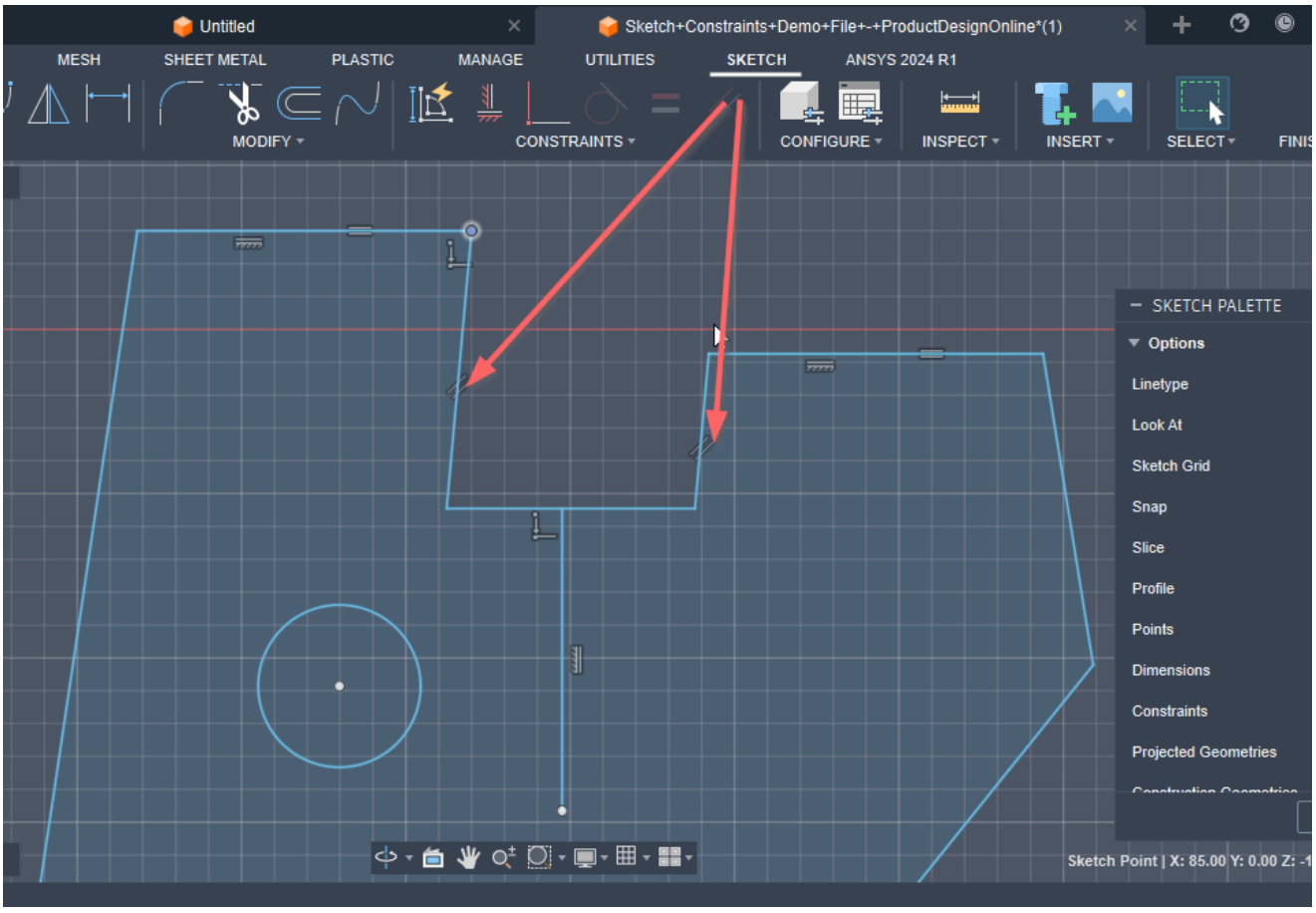


Now they stay equal. Stretch one → the other also stretches.

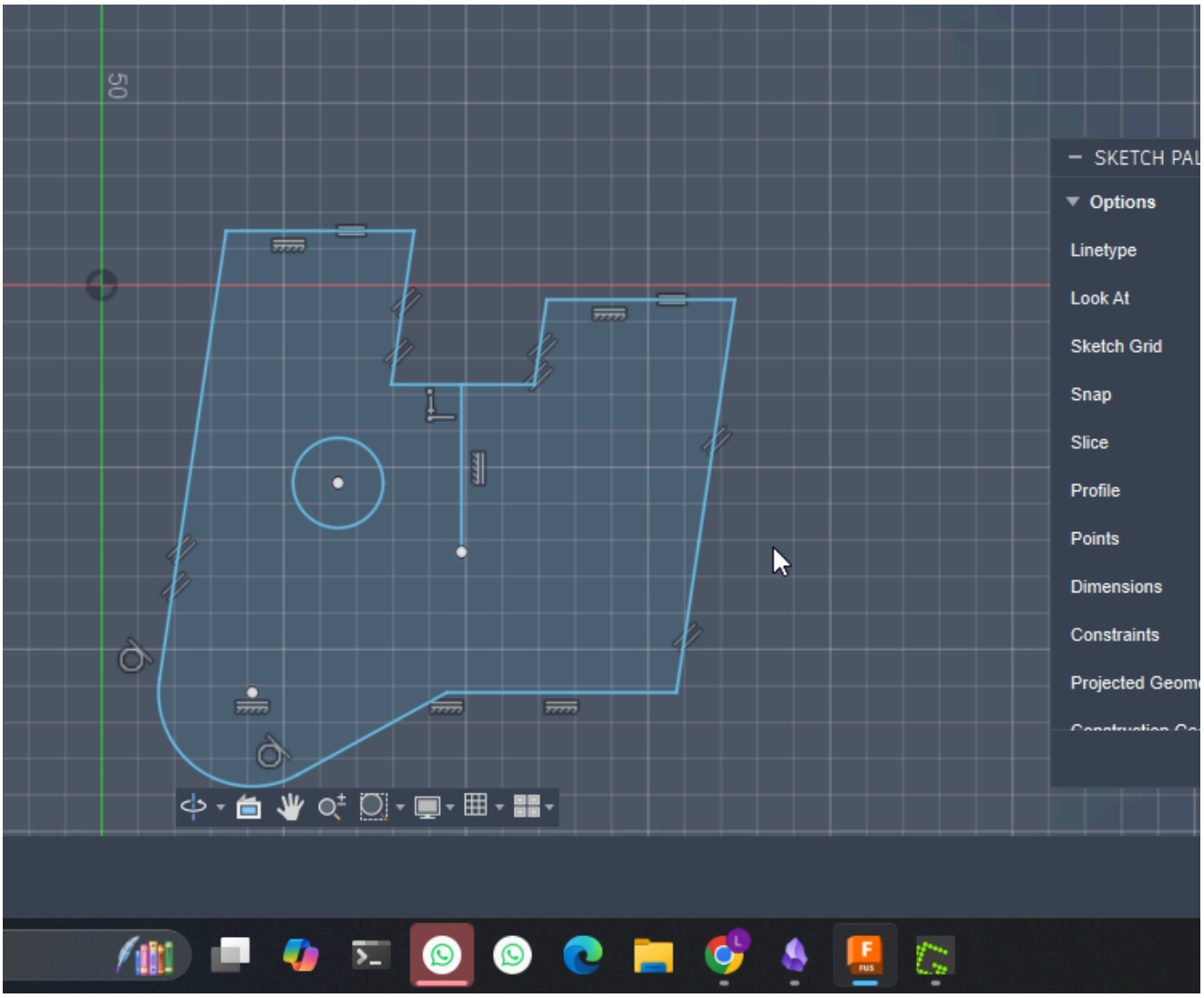


Result.

Parallel Constraint

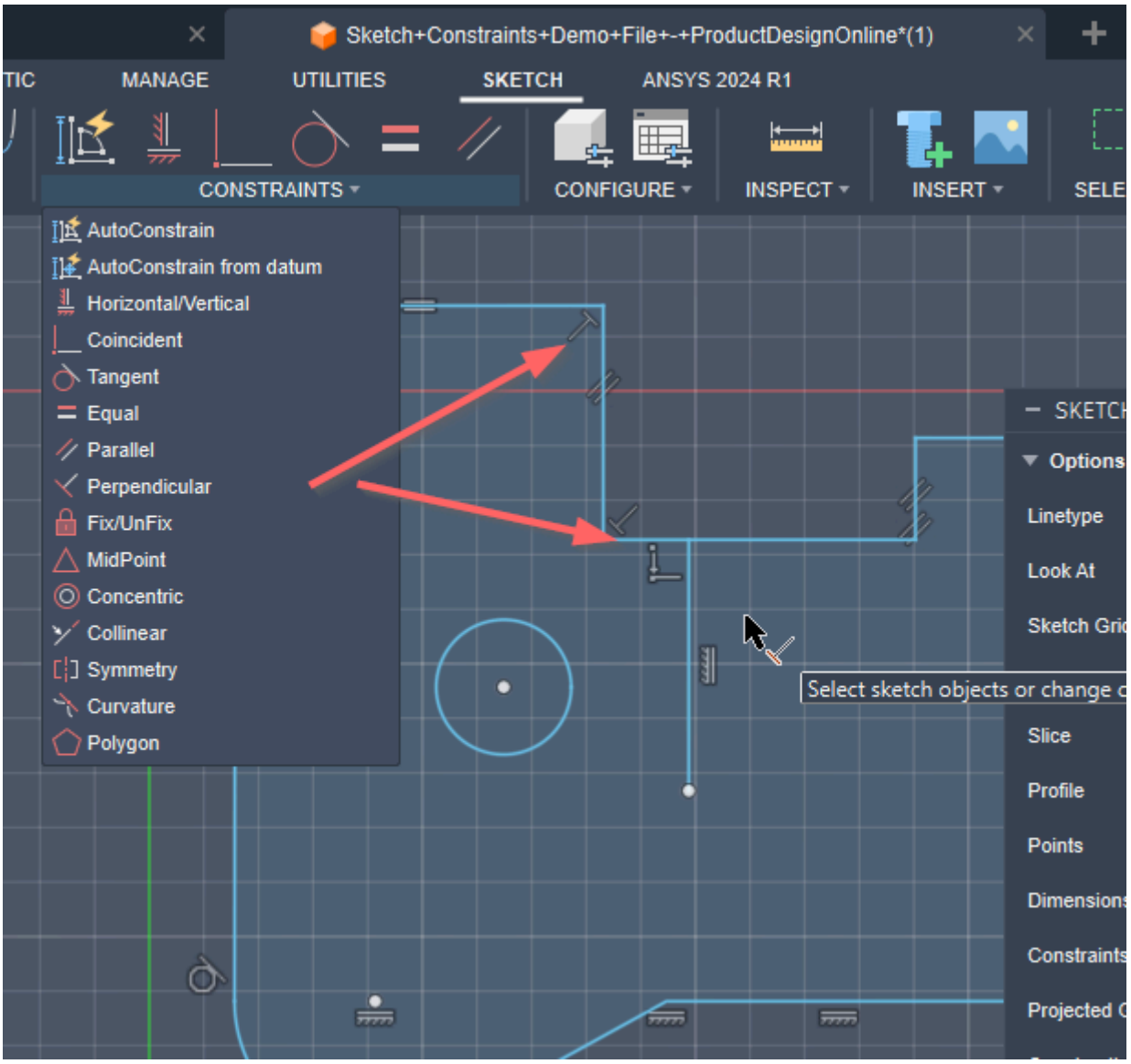


Parallel command.



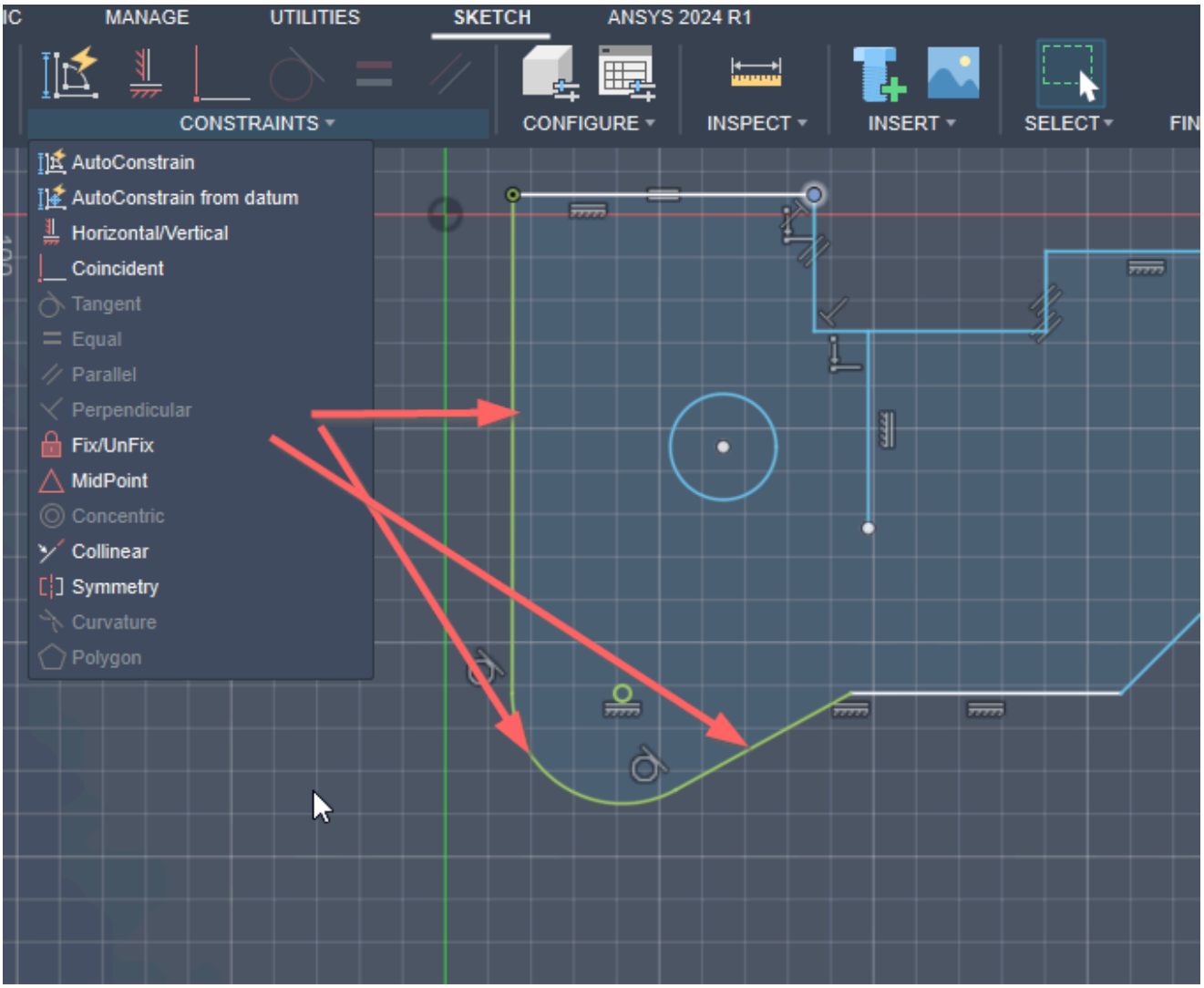
Four sides made parallel.

Perpendicular Constraint



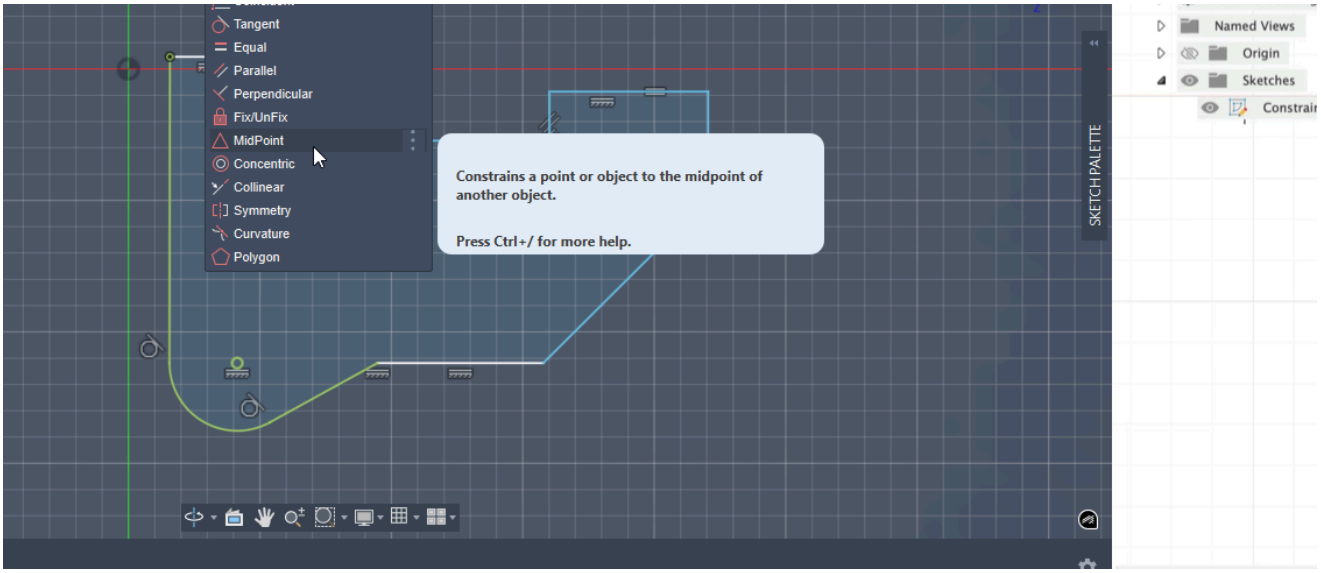
Perpendicular tool.

Fix / Unfix Constraint

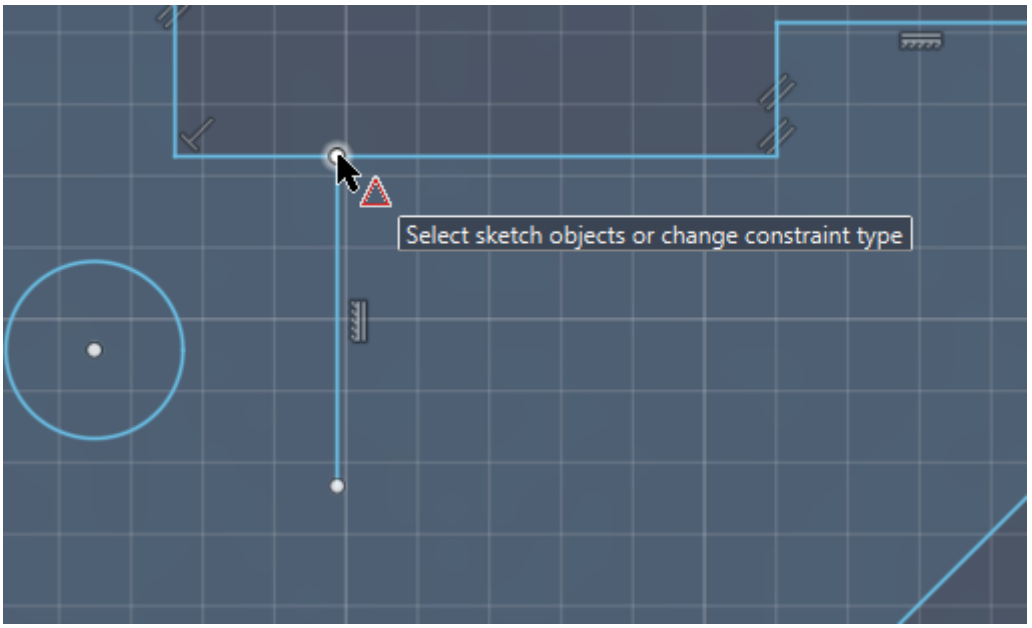


Locks the size and location of a point or object.

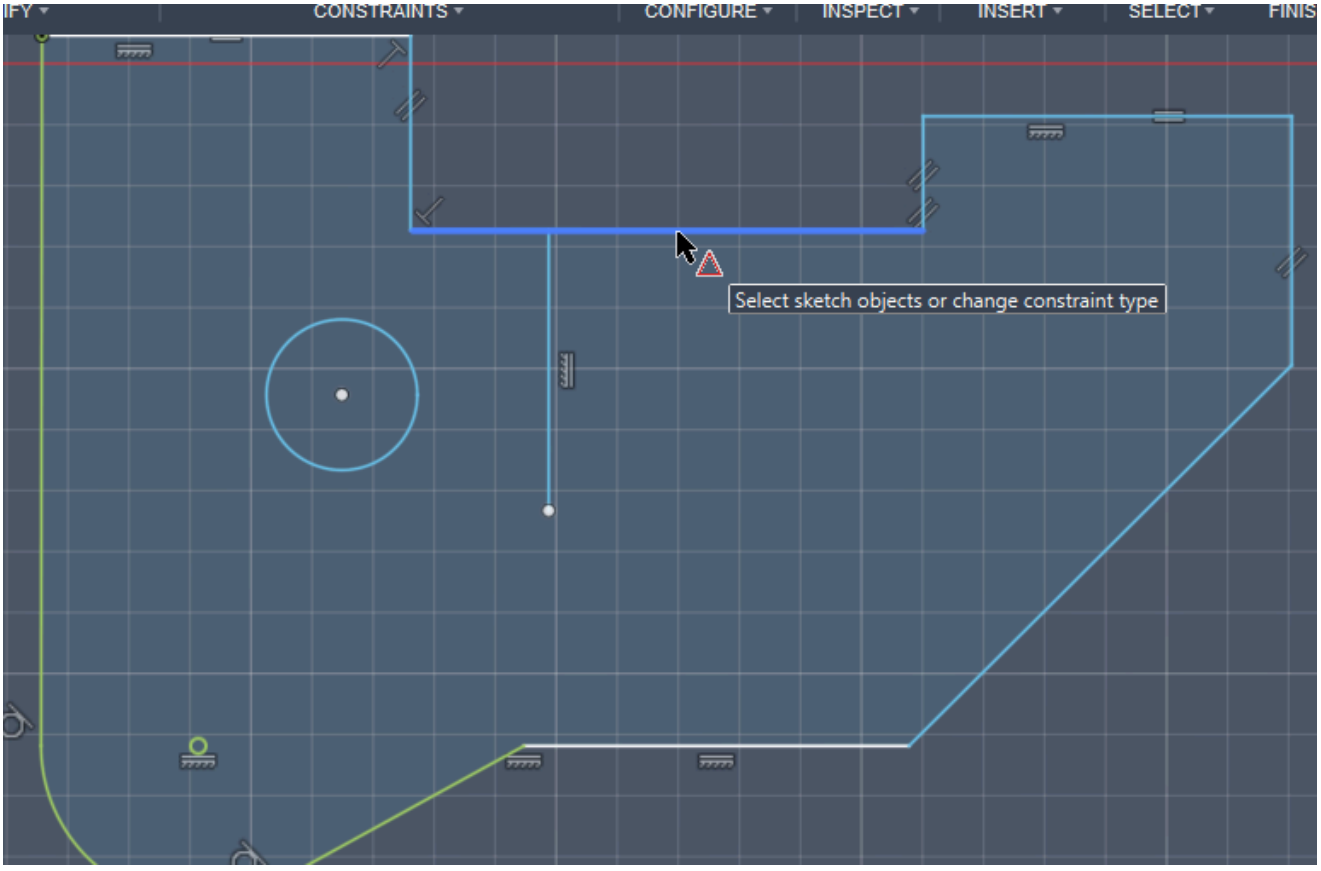
Midpoint Constraint



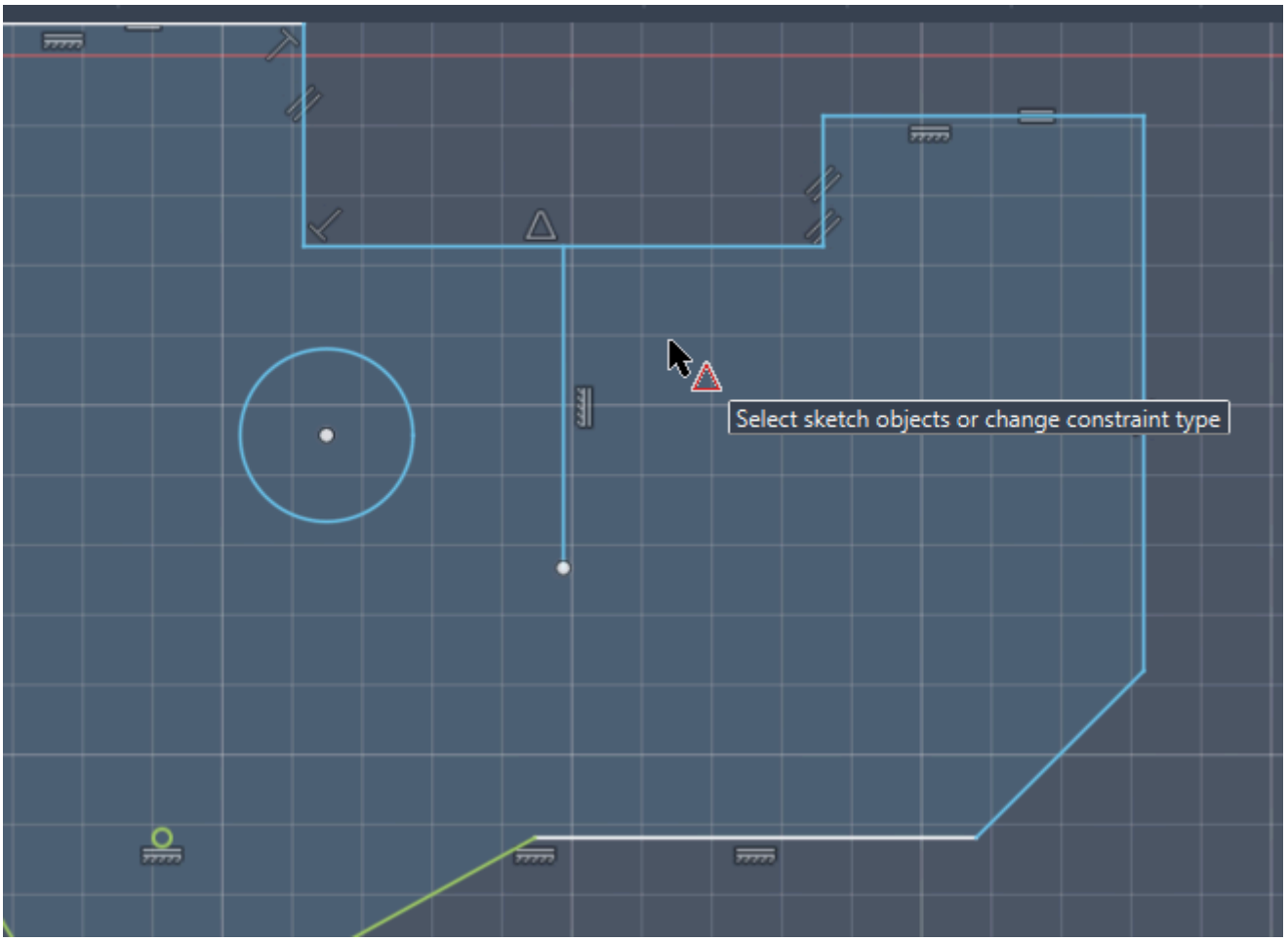
Midpoint tool.



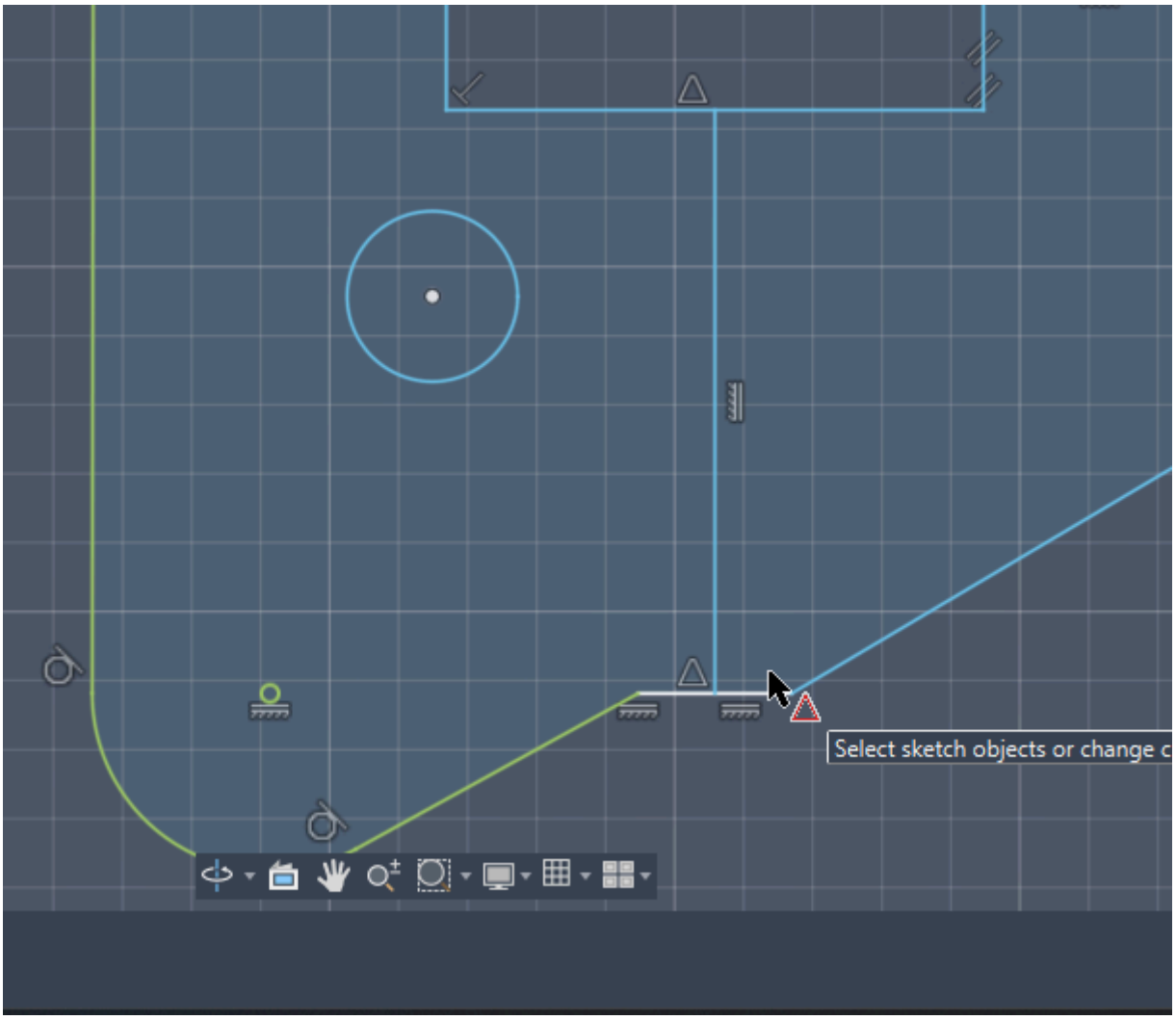
Select this vertex…



Then this line.

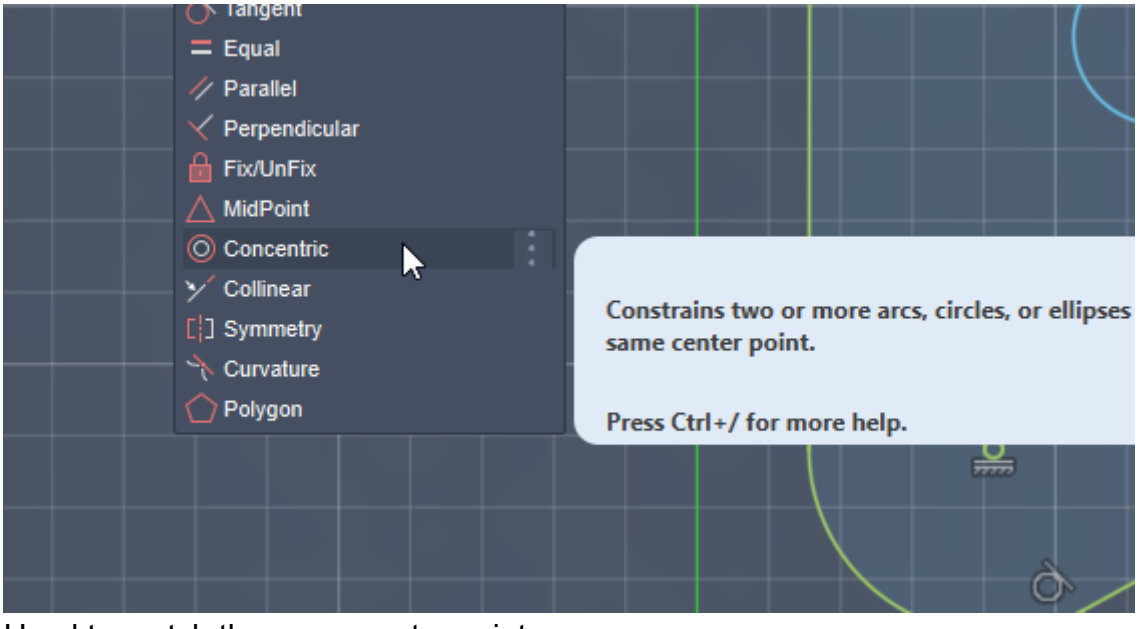


Result.

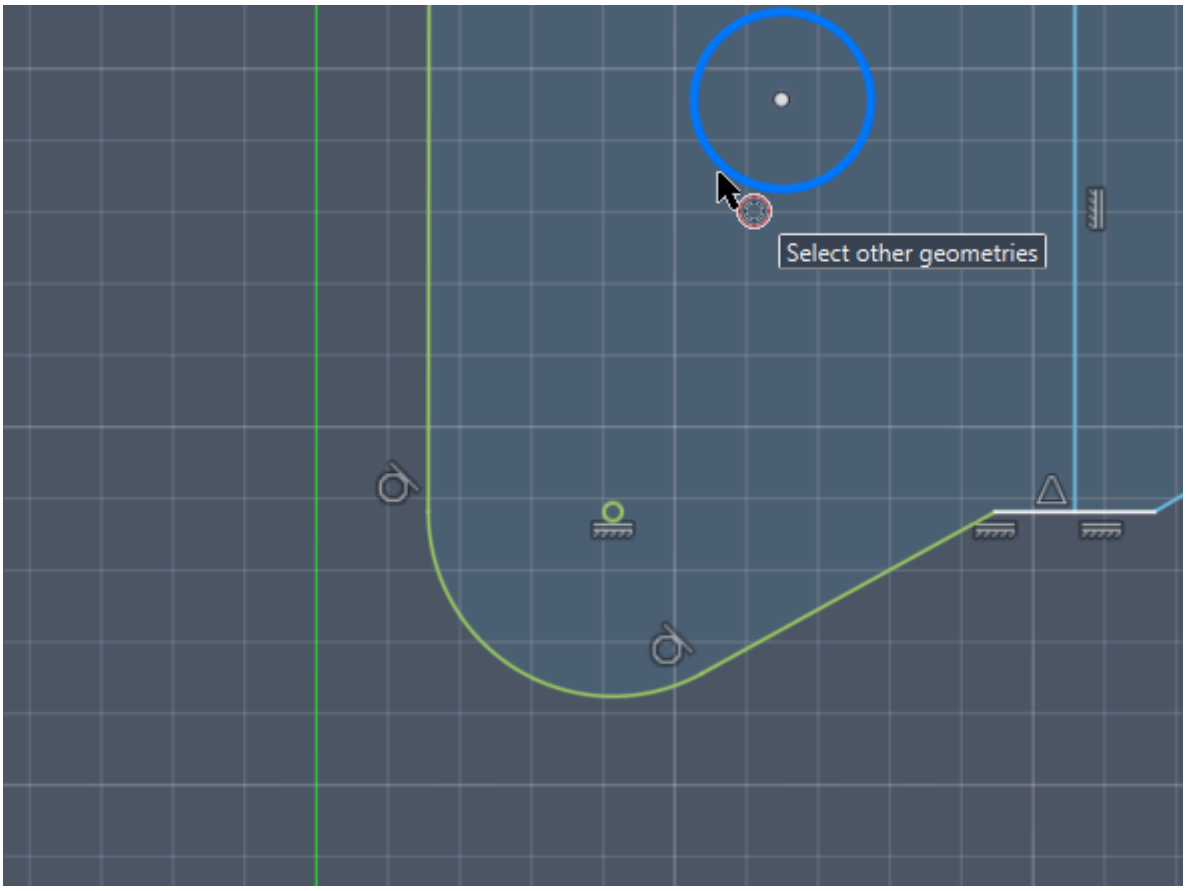


Same for the other vertex.

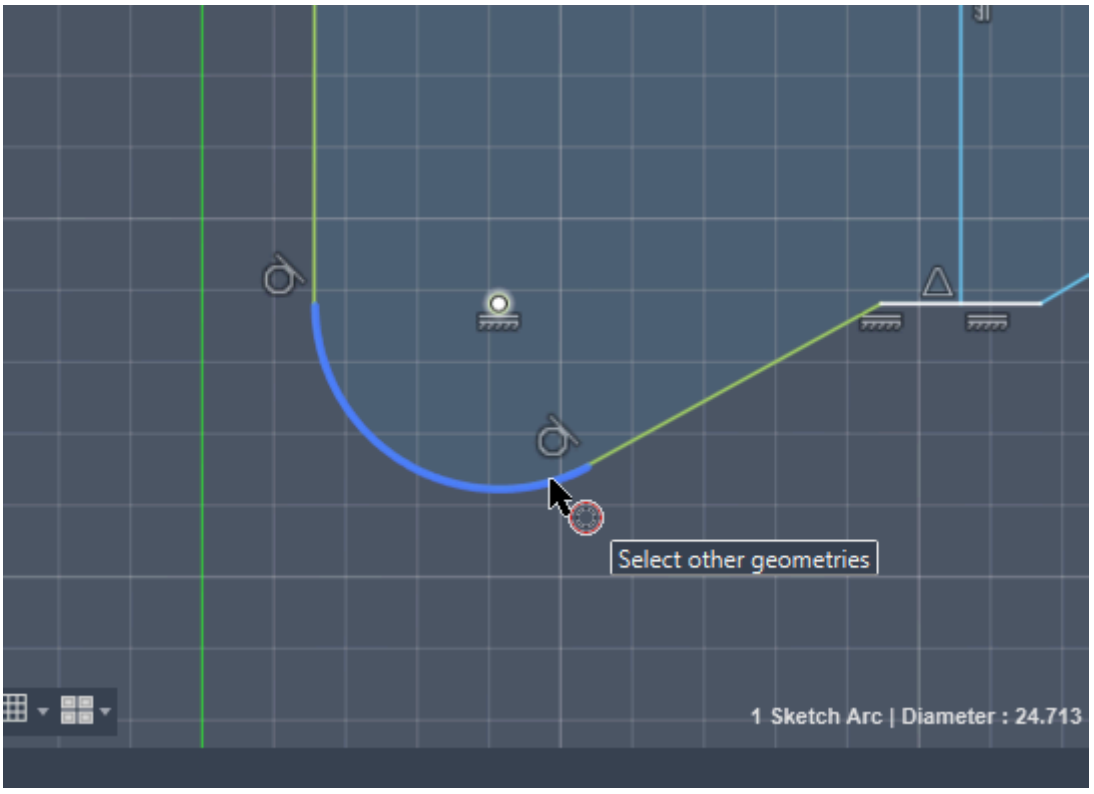
Concentric Constraint



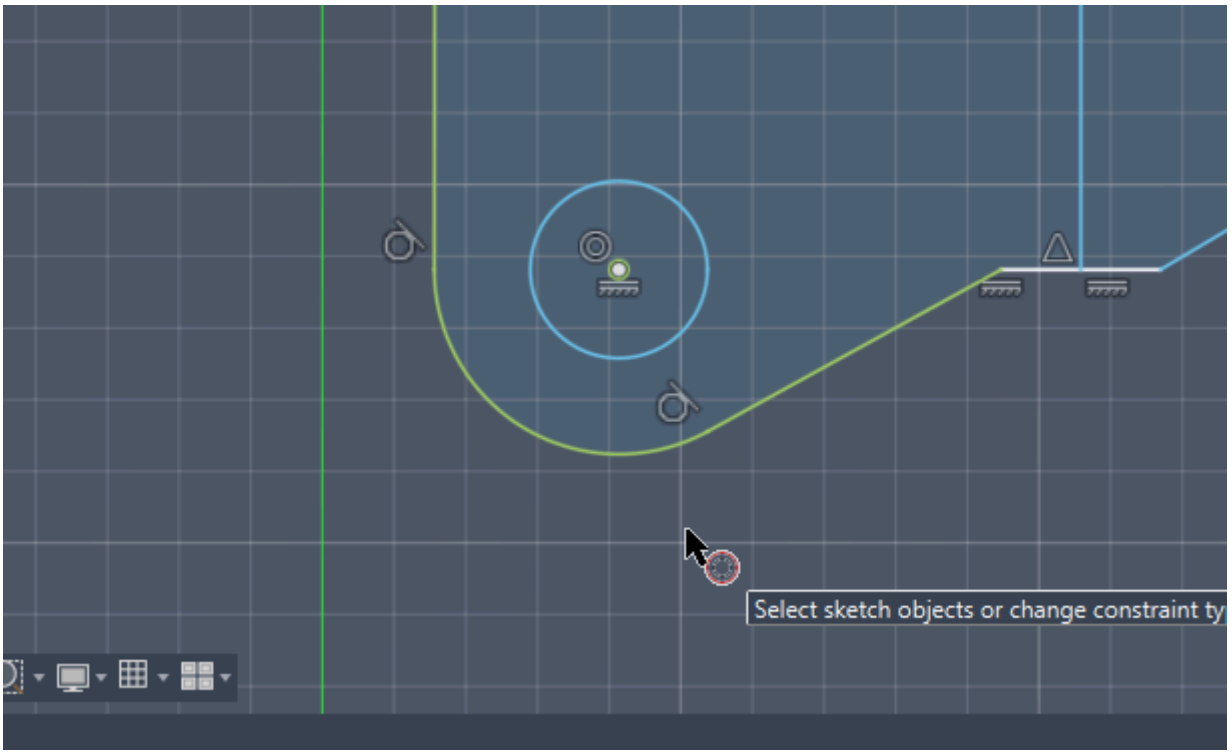
Used to match the same center point.



Select this…

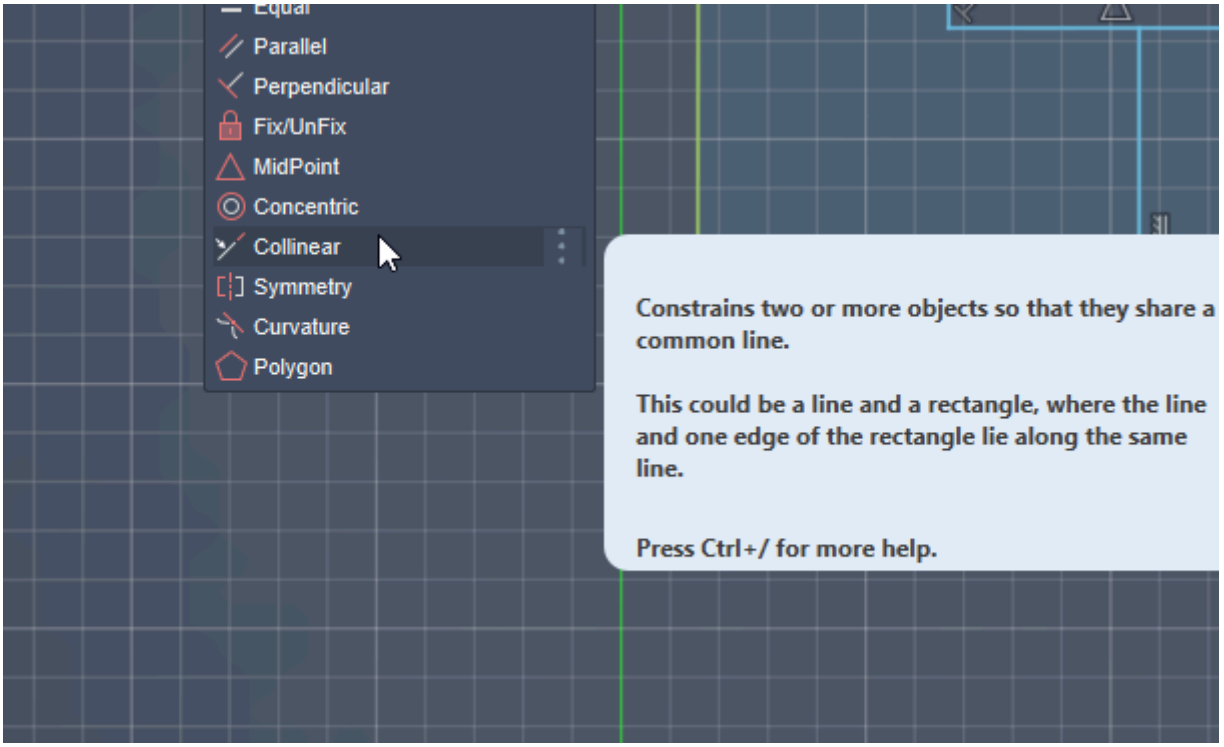


Then this…

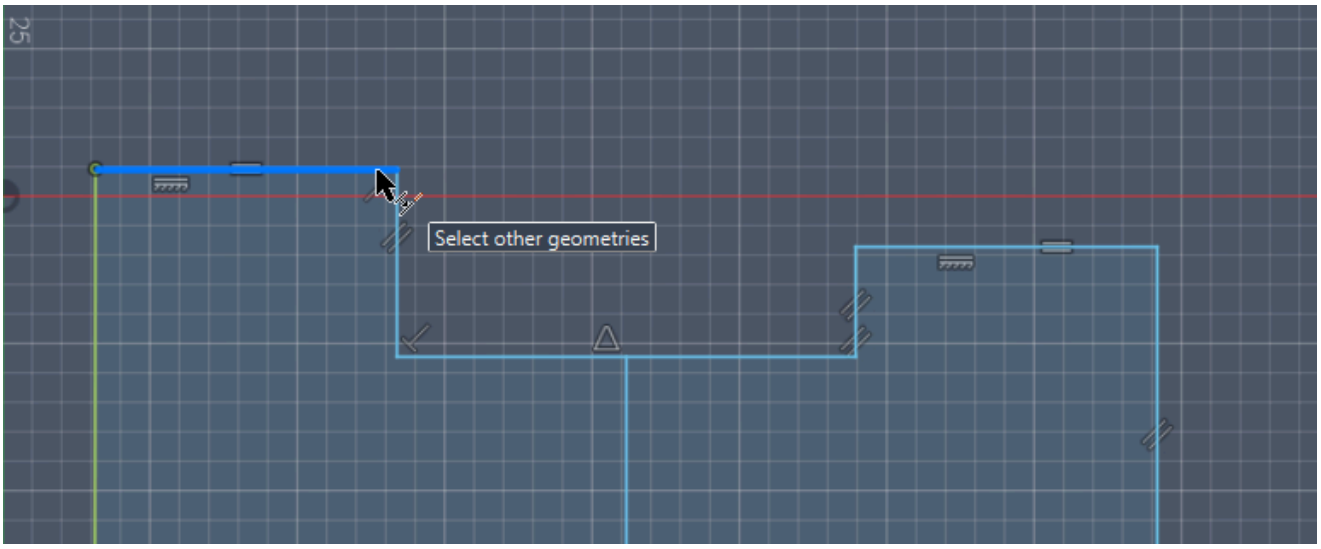


Centers match.

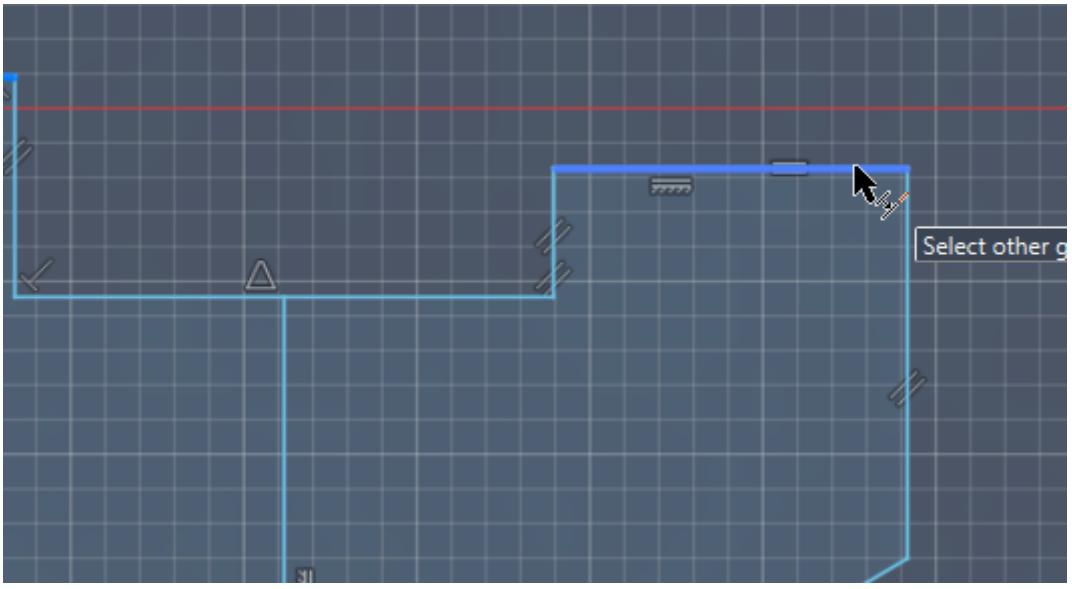
Collinear Constraint



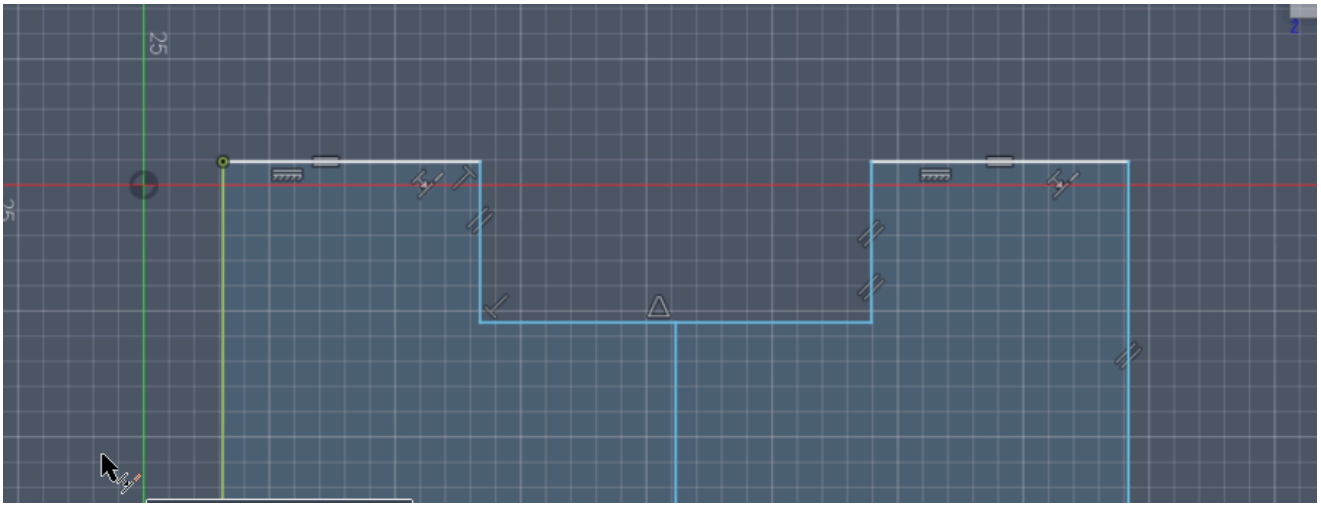
Collinear tool.



Select this…

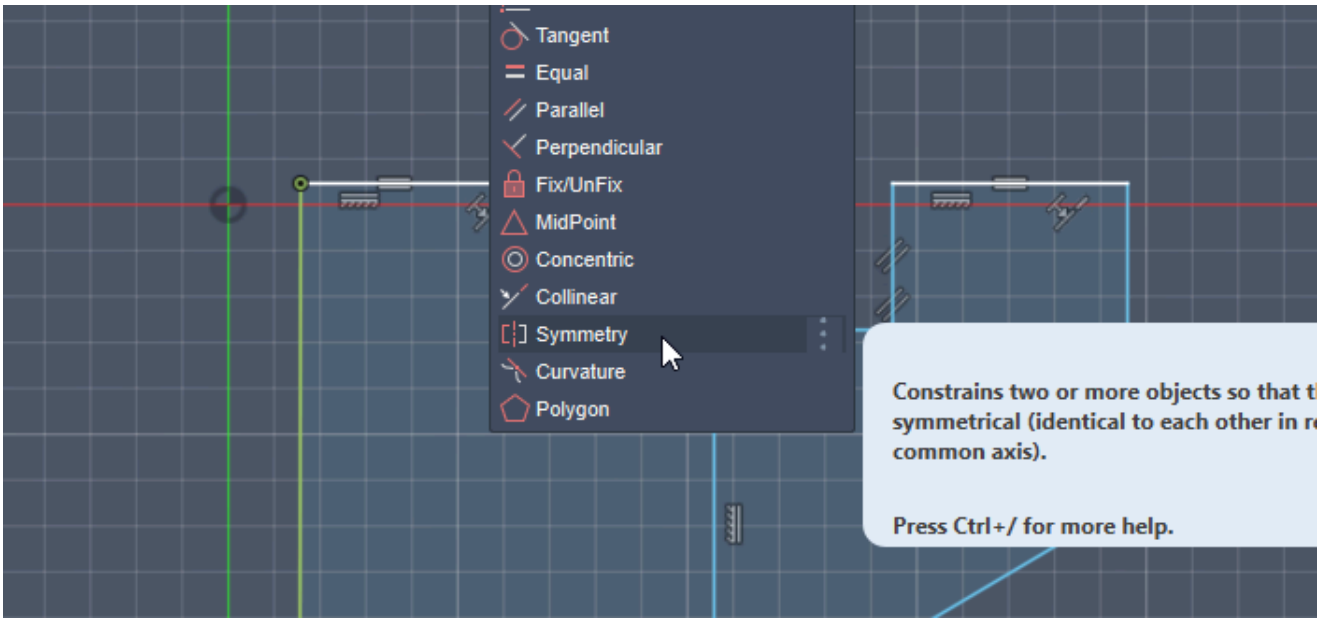


Then this…

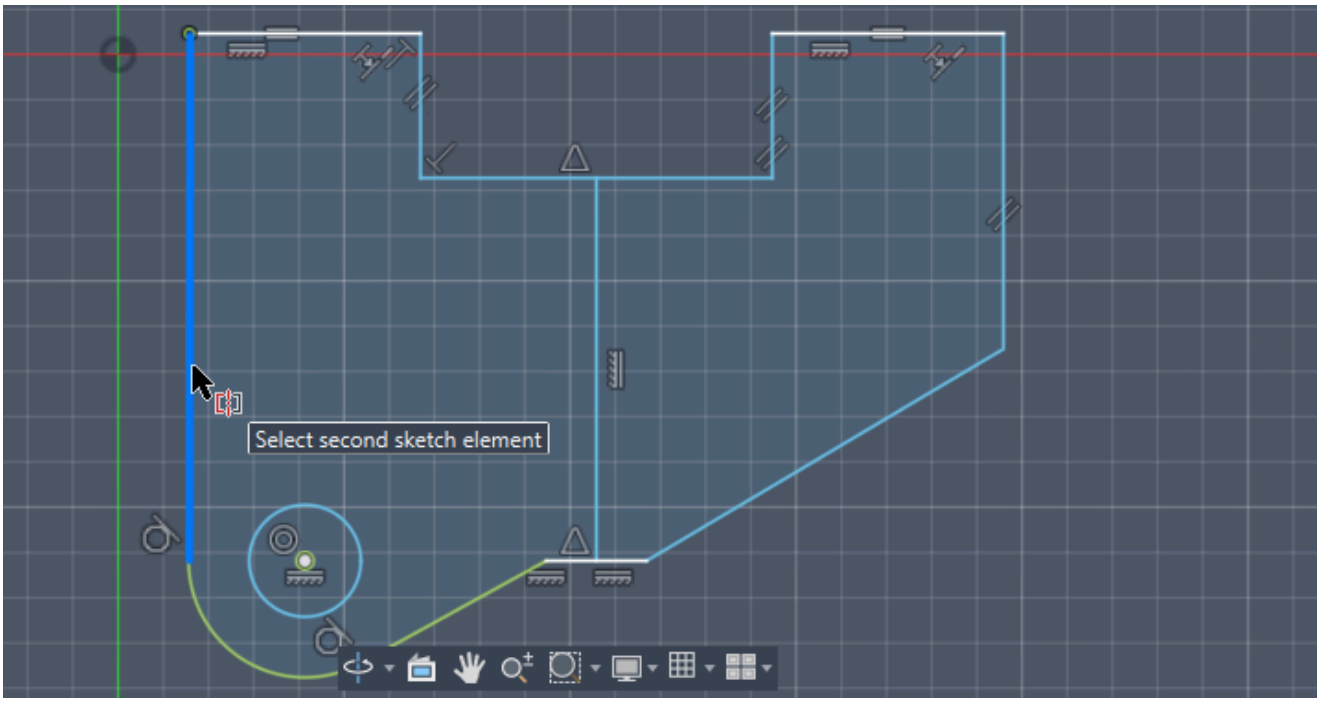


Done.

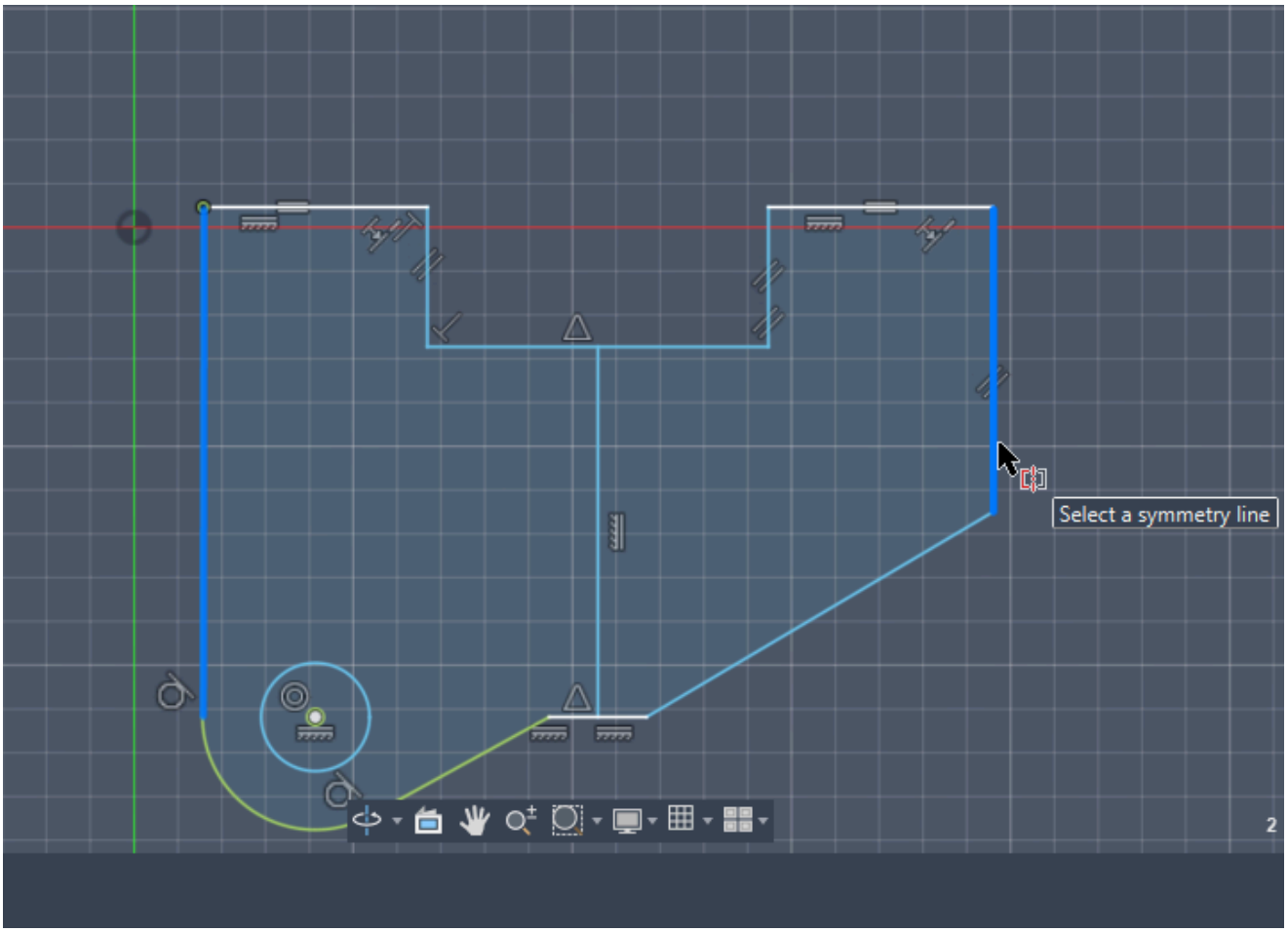
Symmetry Constraint



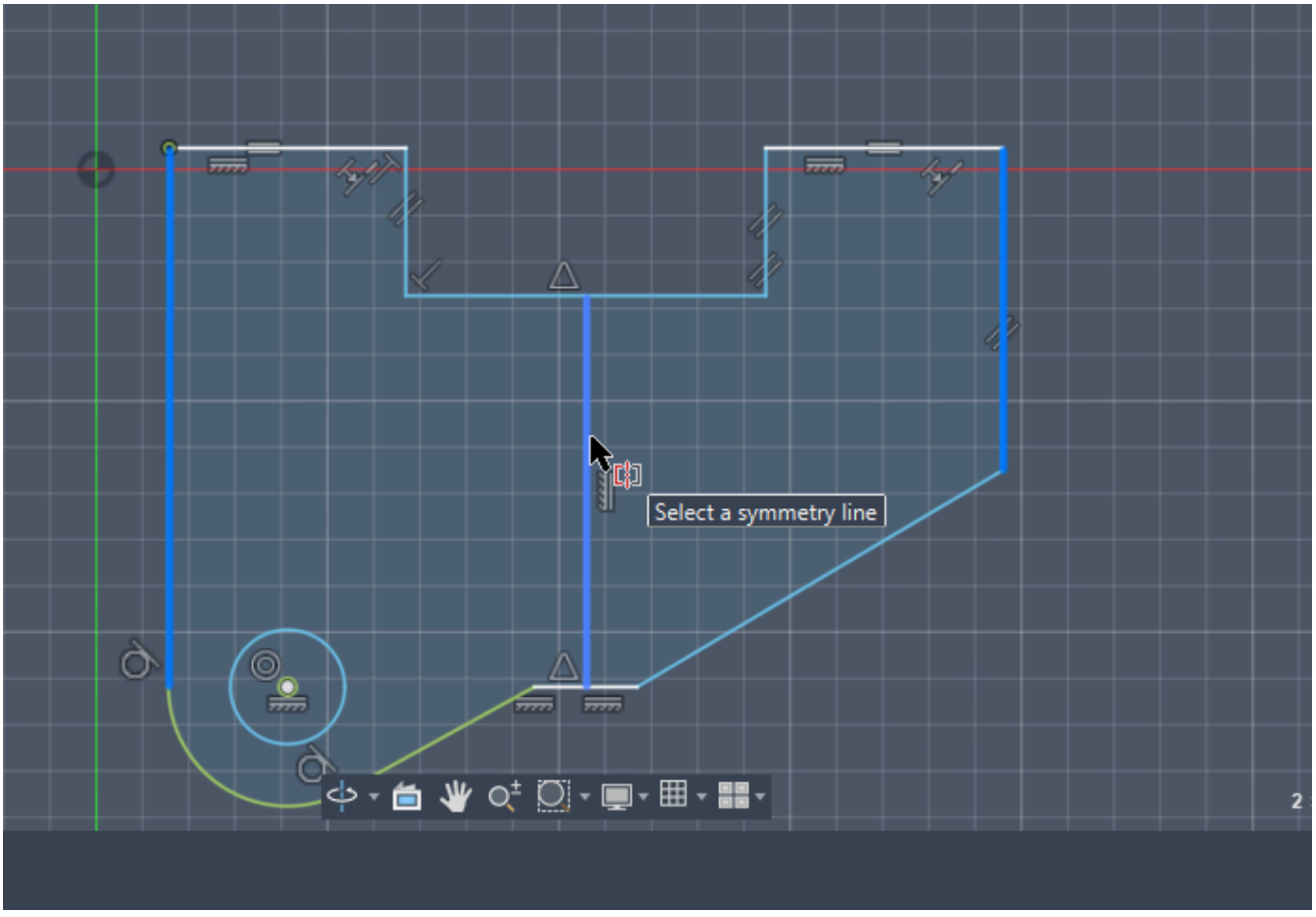
Select first line.



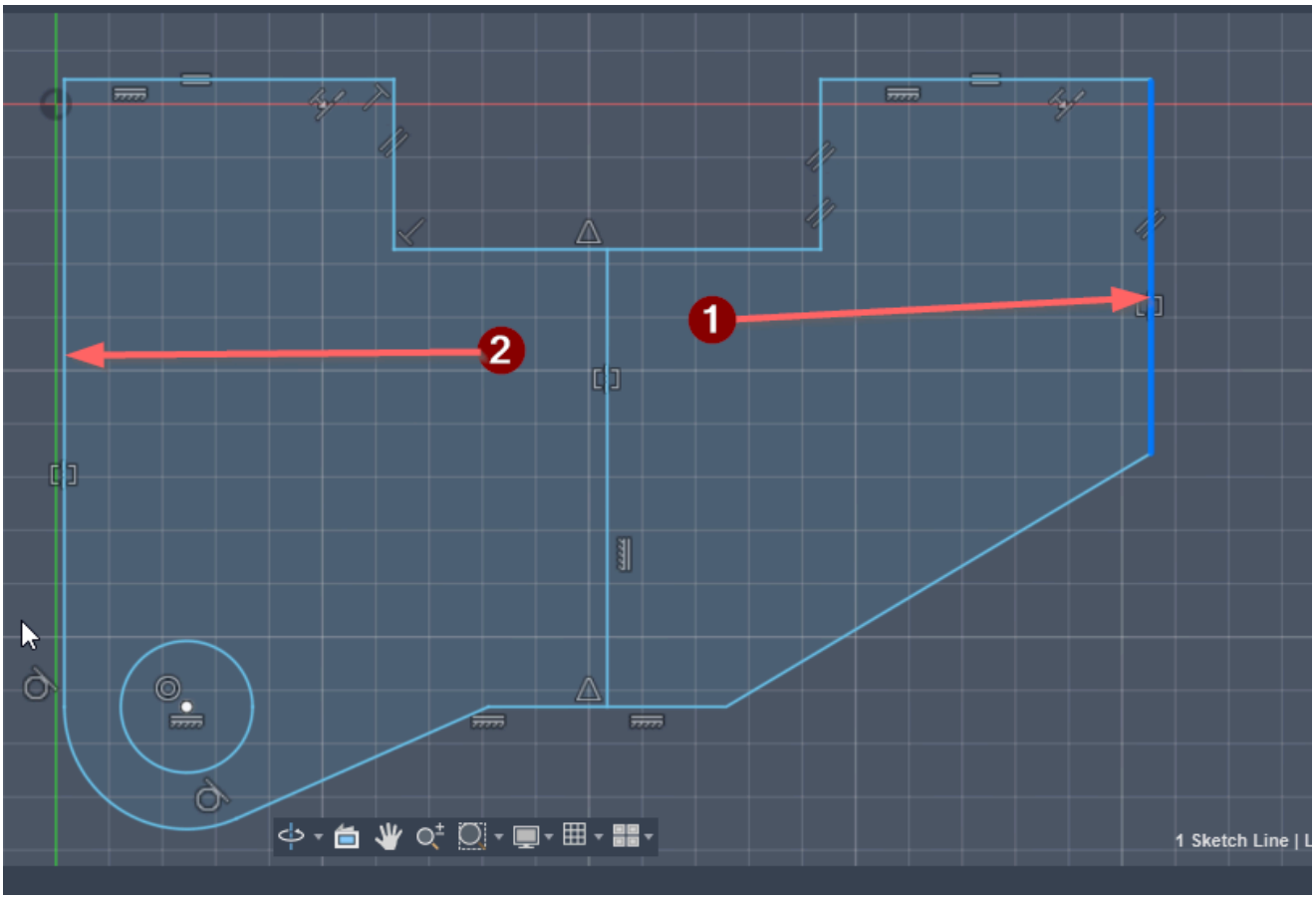
Select second line.



Select center line.

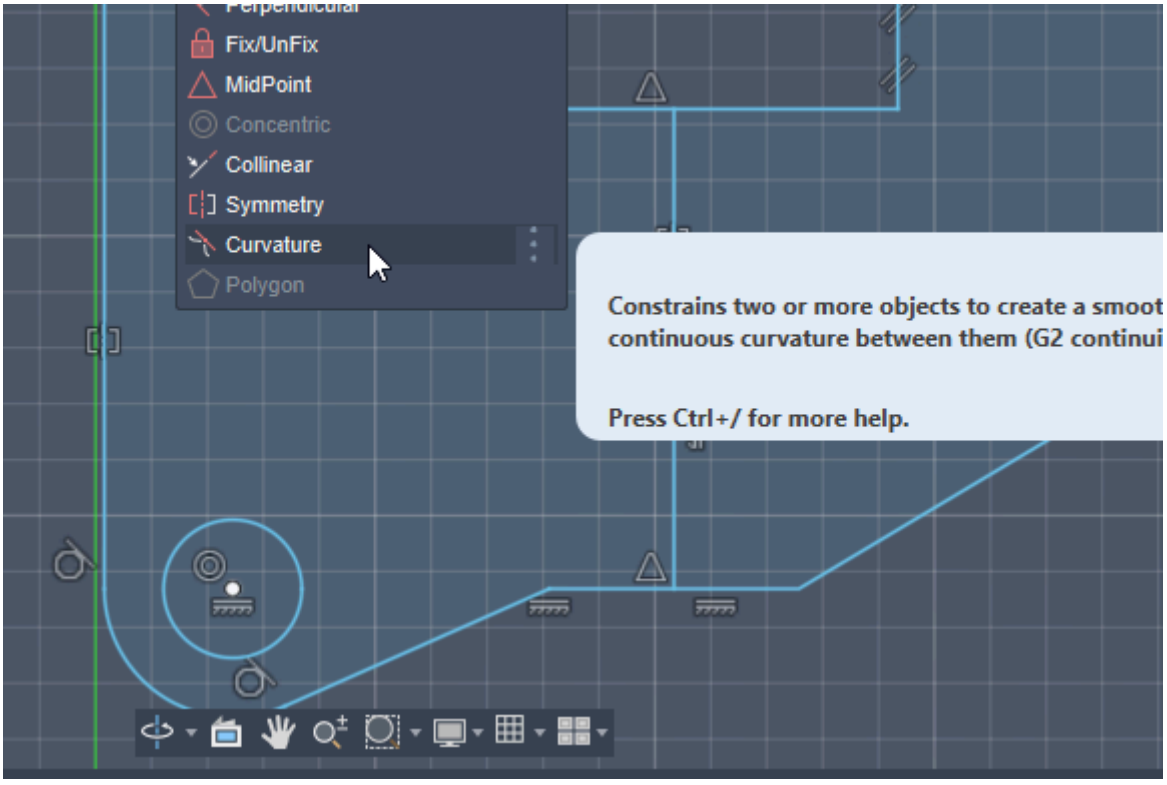


If you move one, the other also moves.



Symmetry applied.

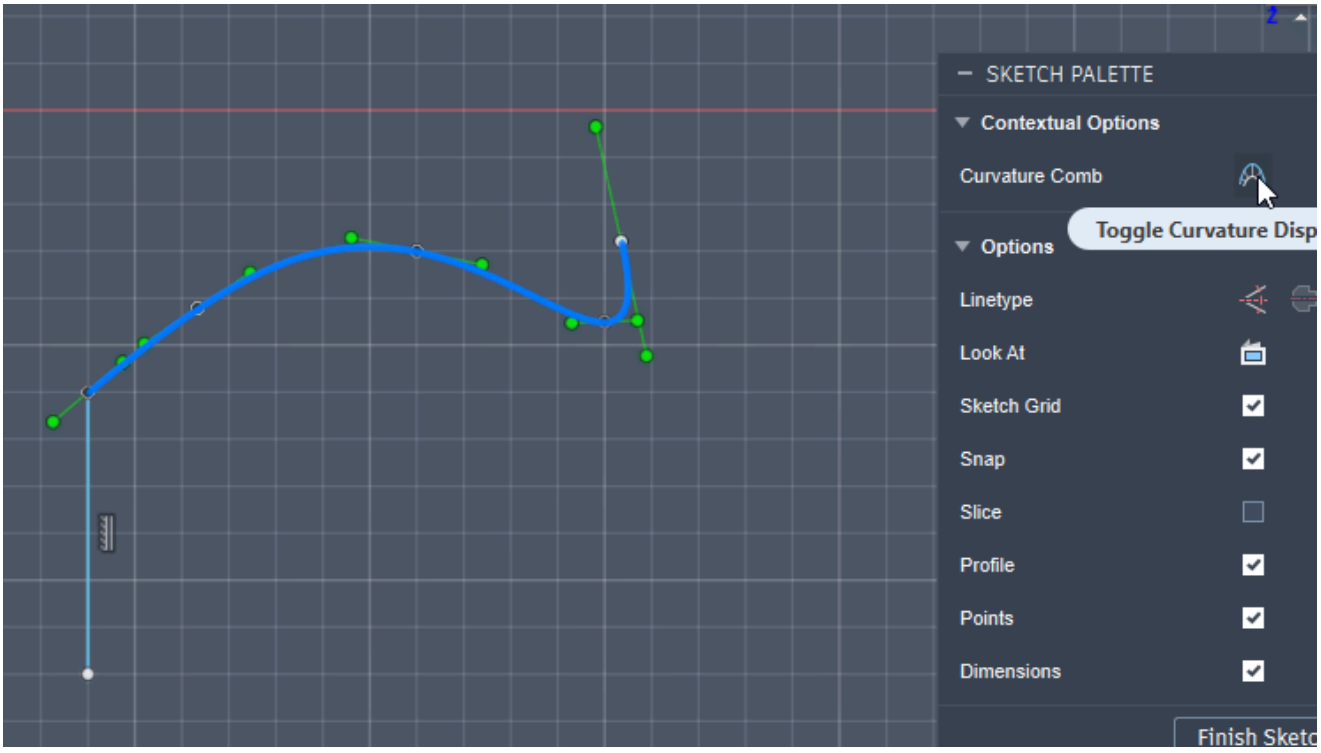
Curvature Constraint



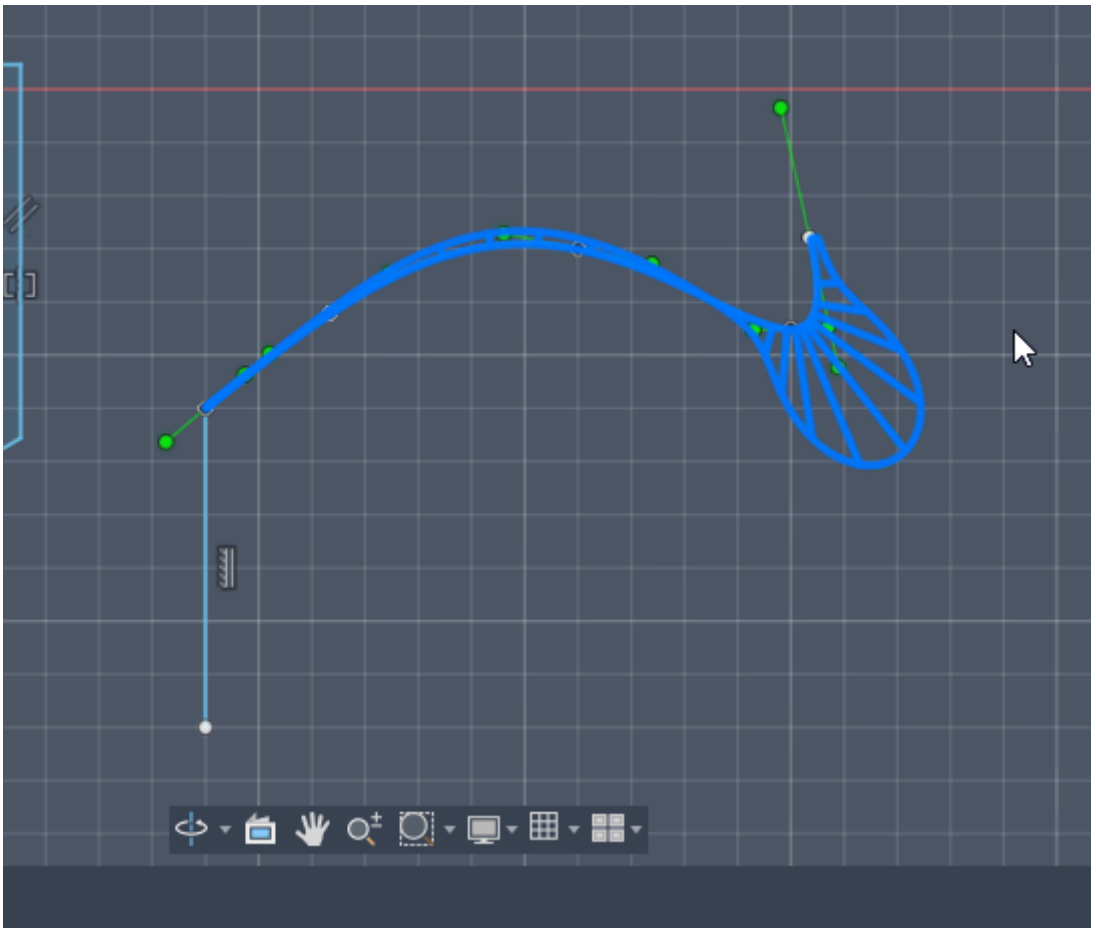
Curvature tool.



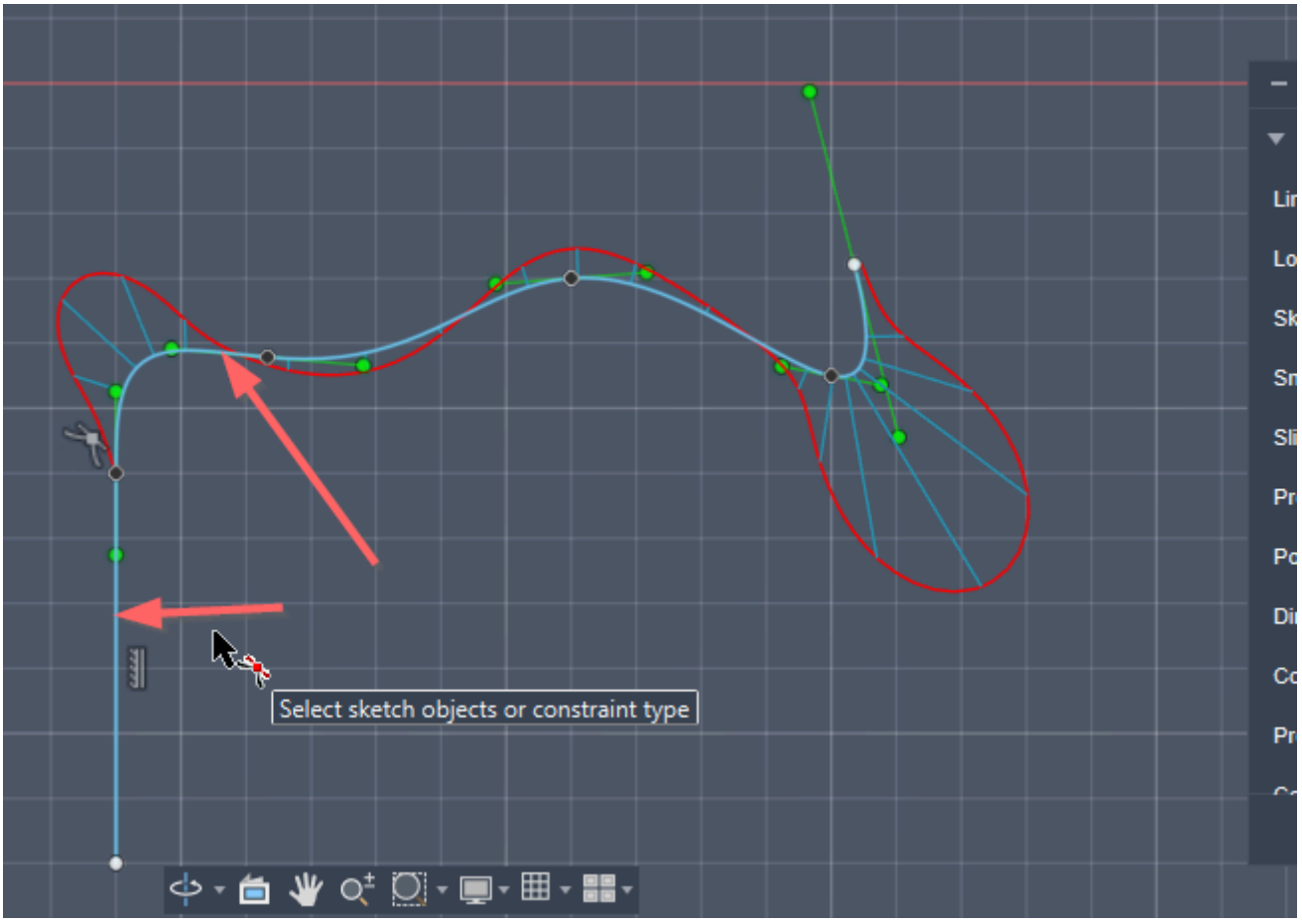
Line (1) and spline (2).



Select fit-point spline, then curvature constraint.

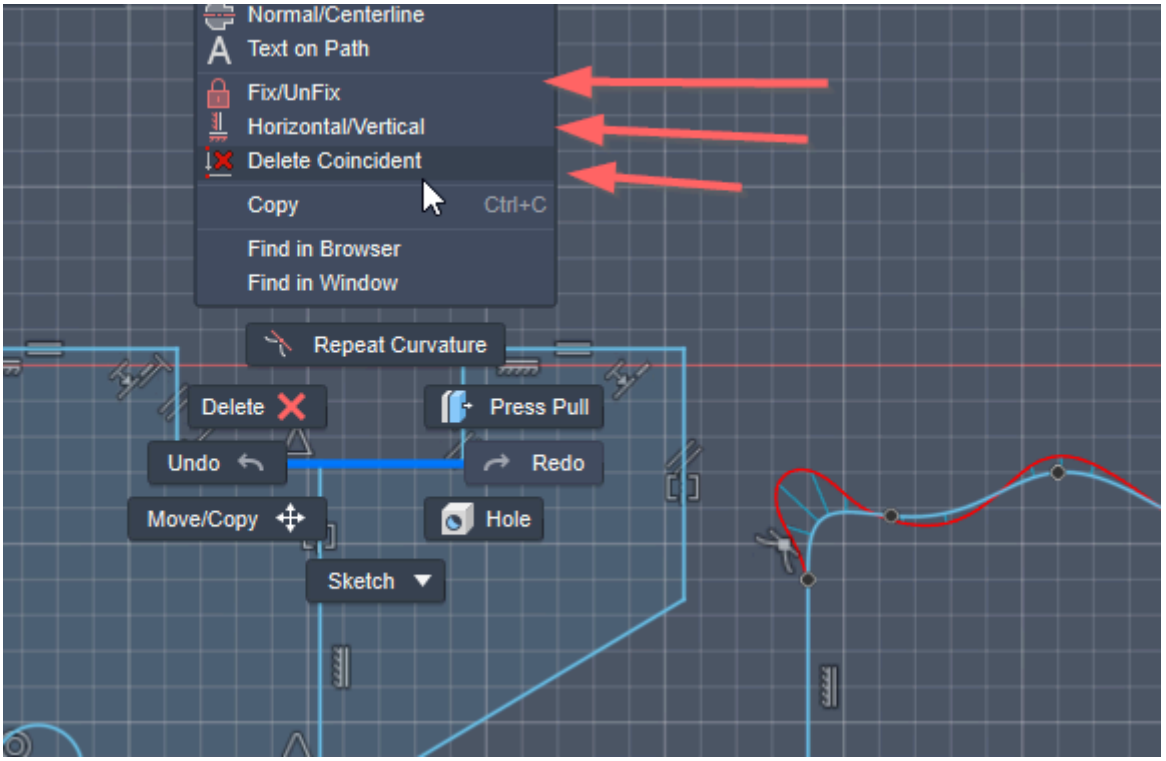


Then apply curvature.



Smooth geometry result.

More Tips

1. 

   Right-clicking shows all relevant sketch constraints.
2. Press **Shift** to select multiple sketch objects.
3. Press **Ctrl** while sketching to disable automatic constraints.